

p. 117 [source PDF p. 503]

[503]

**On the Surfaces each the Locus of the Vertex of a Cone which Passes through  $m$  Given Points and Touches  $6 - m$  Given Lines.**

The factor operated upon by

$$\Delta = X\delta_x + Y\delta_y + Z\delta_z + W\delta_w$$

reduces itself for  $z = 0, w = 0$  to a multiple of  $W$ . Considering the factor in the form of a determinant, the result of the operation is

$$\begin{vmatrix} X & Y & Z & W \\ \sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{a\beta}} & x_a & y_a & z_a \\ \sqrt{\mathfrak{p}_{b\alpha}\mathfrak{p}_{b\beta}} & x_b & y_b & z_b \\ \sqrt{\mathfrak{p}_{c\alpha}\mathfrak{p}_{c\beta}} & x_c & y_c & z_c \\ \sqrt{\mathfrak{p}_{d\alpha}\mathfrak{p}_{d\beta}} & x_d & y_d & z_d, w_d \end{vmatrix} + \begin{vmatrix} 0 & x & y & . & . \\ \Delta\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{a\beta}} & x_a & y_a & z_a & . \\ \Delta\sqrt{\mathfrak{p}_{b\alpha}\mathfrak{p}_{b\beta}} & x_b & y_b & z_b & . \\ \Delta\sqrt{\mathfrak{p}_{c\alpha}\mathfrak{p}_{c\beta}} & x_c & y_c & z_c & . \\ \Delta\sqrt{\mathfrak{p}_{d\alpha}\mathfrak{p}_{d\beta}} & x_d & y_d & z_d & w_d \end{vmatrix}.$$

The first term is

$$\begin{vmatrix} X & Y & Z & W \\ z_a\sqrt{I_\alpha I_\beta} & x_a & y_a & z_a \\ z_b\sqrt{I_\alpha I_\beta} & x_b & y_b & z_b \\ z_c\sqrt{I_\alpha I_\beta} & x_c & y_c & z_c \\ \sqrt{\mathfrak{p}_{d\alpha}\mathfrak{p}_{d\beta}} & x_d & y_d & z_d, w_d \end{vmatrix},$$

where the first column may be replaced by

$$\begin{aligned} & -Z\sqrt{I_\alpha I_\beta} \\ & \vdots \\ & \sqrt{\mathfrak{p}_{d\alpha}\mathfrak{p}_{d\beta}} - z_d\sqrt{I_\alpha I_\beta} \end{aligned}$$

and the term in question thus becomes

$$\{w_d Z\sqrt{I_\alpha I_\beta} + W(-z_d\sqrt{I_\alpha I_\beta} + \sqrt{\mathfrak{p}_{d\alpha}\mathfrak{p}_{d\beta}})\} abc,$$

if for shortness

$$\begin{vmatrix} x_a & y_a & z_a \\ x_b & y_b & z_b \\ x_c & y_c & z_c \end{vmatrix} = abc.$$

As regards the second term, we have

$$\Delta\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{a\beta}} = \frac{\mathfrak{p}_{a\alpha}\Delta\mathfrak{p}_{a\beta}\mathfrak{p}_{a\beta}\Delta\mathfrak{p}_{a\alpha}}{2\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{a\beta}}},$$

which is

$$= \frac{I_\alpha\Delta\mathfrak{p}_{a\beta} + I_\beta\Delta\mathfrak{p}_{a\alpha}}{2\sqrt{I_\alpha I_\beta}}.$$

But

$$\begin{aligned} \mathfrak{p}_{a\alpha} &= x(-g_\alpha z_a) + y(f_\alpha z_a) + z(g_\alpha x_a - f_\alpha y_a) + w(a_\alpha x_a - b_\alpha y_a - c_\alpha z_a) \\ &= x_a(g_\alpha z - a_\alpha w) + y_a(-f_\alpha z - b_\alpha w) + z_a(-g_\alpha x + f_\alpha y - c_\alpha w); \end{aligned}$$

and thence

$$\Delta\mathfrak{p}_{a\alpha} = x_a(g_\alpha Z - a_\alpha W) + y_a(-f_\alpha Z - b_\alpha W) + z_a(-g_\alpha X + f_\alpha Y - c_\alpha W).$$

{Surface  $abcd\alpha\beta$ .}

p. 118 [source PDF p. 503]

With the like formula for  $\Delta \mathfrak{p}_{a\beta}$ ; hence

$$\frac{I_\alpha \Delta \mathfrak{p}_{a\beta} + I_\beta \Delta \mathfrak{p}_{a\alpha}}{2\sqrt{I_\alpha I_\beta}} = Ax_a + By_a + Cz_a,$$

where

$$\begin{aligned} A &= \frac{1}{2\sqrt{I_\alpha I_\beta}} \{I_\beta(g_\alpha Z - a_\alpha W) + I_\alpha(g_\beta Z - a_\beta W)\}, \\ B &= \frac{1}{2\sqrt{I_\alpha I_\beta}} \{I_\beta(-f_\alpha Z - b_\alpha W) + I_\alpha(-f_\beta Z - b_\beta W)\}, \\ C &= \frac{1}{2\sqrt{I_\alpha I_\beta}} \{I_\beta(-g_\alpha X + f_\alpha Y - c_\alpha W) + I_\alpha(-g_\beta X + f_\beta Y - c_\beta W)\}. \end{aligned}$$

The term in question is thus

$$\begin{vmatrix} 0 & x & y & . & . \\ Ax_a + By_a + Cz_a & x_a & y_a & z_a & . \\ Ax_b + By_b + Cz_b & x_b & y_b & z_b & . \\ Ax_c + By_c + Cz_c & x_c & y_c & z_c & . \\ \Delta\sqrt{\mathfrak{p}_{d\alpha}\mathfrak{p}_{d\beta}} & x_d & y_d & z_d & w_d \end{vmatrix}.$$

Viz. replacing the first column by

$$\begin{aligned} &-Ax - By \\ &\vdots \\ &\Delta\sqrt{\mathfrak{p}_{d\alpha}\mathfrak{p}_{d\beta}} - Ax_d - By_d - Cz_d \end{aligned}$$

this is

$$= (Ax + By)w_d abc;$$

and we have

$$\begin{aligned} Ax + By &= \frac{1}{2\sqrt{I_\alpha I_\beta}} [\{I_\beta(g_\alpha x - f_\alpha y) + I_\alpha(g_\beta x - f_\beta y)\}Z \\ &\quad + \{I_\beta(-a_\alpha x - b_\alpha y) + I_\alpha(-a_\beta x - b_\beta y)\}W] \\ &= \frac{1}{2\sqrt{I_\alpha I_\beta}} (-2I_\alpha I_\beta Z - MW), \end{aligned}$$

if for shortness

$$M = (-g_\beta x + f_\beta y)(a_\alpha x + b_\alpha y) + (-g_\alpha x + f_\alpha y)(a_\beta x + b_\beta y).$$

Viz. the whole term is

$$w_d \left\{ -\sqrt{I_\alpha I_\beta} Z - \frac{\frac{1}{2}M}{\sqrt{I_\alpha I_\beta}} W \right\} abc.$$

Hence the first and second terms together are

$$= W \left\{ -z_d \sqrt{I_\alpha I_\beta} + \sqrt{\mathfrak{p}_{d\alpha}\mathfrak{p}_{d\beta}} - \frac{\frac{1}{2}M}{\sqrt{I_\alpha I_\beta}} w_d \right\} abc;$$

viz. this is a multiple of  $W$ , which was the theorem to be proved.

{Surface  $abcd\alpha\beta$ .}

p. 119 [source PDF p. 503]

**Surface  $abc\alpha\beta\gamma$ .**

32. The equation is

$$\text{Norm} \begin{vmatrix} \sqrt{\mathfrak{p}_{a\alpha}} & \sqrt{\mathfrak{p}_{b\alpha}} & \sqrt{\mathfrak{p}_{c\alpha}} \\ \sqrt{\mathfrak{p}_{a\beta}} & \sqrt{\mathfrak{p}_{b\beta}} & \sqrt{\mathfrak{p}_{c\beta}} \\ \sqrt{\mathfrak{p}_{a\gamma}} & \sqrt{\mathfrak{p}_{b\gamma}} & \sqrt{\mathfrak{p}_{c\gamma}} \end{vmatrix} = 0,$$

where the norm is a product of 16 factors, each of the order  $\frac{3}{2}$ . As before,  $\mathfrak{p}_{a\alpha} = 0$  is the equation of the plane through the point  $a$  and the line  $\alpha$ ; viz.  $\mathfrak{p}_{a\alpha}$  has the value already mentioned.

33. *Investigation.* In the projection, the equation of the conic touching the projections of the lines  $\alpha, \beta, \gamma$  is

$$A\sqrt{P_\alpha X + Q_\alpha Y + R_\alpha Z} + B\sqrt{P_\beta X + Q_\beta Y + R_\beta Z} + C\sqrt{P_\gamma X + Q_\gamma Y + R_\gamma Z} = 0;$$

and to make this pass through the projection of the point  $a$ , we must write herein  $X : Y : Z = p_a : q_a : r_a$ . As before, we have

$$\begin{aligned} P_\alpha X + Q_\alpha Y + R_\alpha Z &= w_a(xP_\alpha + yQ_\alpha + zR_\alpha) \\ &\quad - w(x_aP_\alpha + y_aQ_\alpha + z_aR_\alpha) \\ &= -w(x_aP_\alpha + y_aQ_\alpha + z_aR_\alpha + w_aS_\alpha) \\ &= -w\mathfrak{p}_{a\alpha}; \end{aligned}$$

and so for the other terms; the equation thus is

$$A\sqrt{\mathfrak{p}_{a\alpha}} + B\sqrt{\mathfrak{p}_{a\beta}} + C\sqrt{\mathfrak{p}_{a\gamma}} = 0;$$

or forming the like equations in regard to the points  $b, c$  respectively, and eliminating, we have a determinant  $= 0$ , and then, taking the norm, we obtain the above-written equation of the surface.

34. *Singularities.* The equation of the surface shows that

- (0) The point  $a$  is 8-conical: for it  $\mathfrak{p}_{a\alpha} = \mathfrak{p}_{a\beta} = \mathfrak{p}_{a\gamma} = 0$ ;  
each factor is  $O^{1/2}$ , and the norm is  $O^8$ .

- (1) The line  $ab$  is 4-tuple.

To prove this, observe that the sixteen factors are obtained by attributing at pleasure the signs  $+$ ,  $-$  to the radicals  $\sqrt{\mathfrak{p}_{b\beta}}, \sqrt{\mathfrak{p}_{c\beta}}, \sqrt{\mathfrak{p}_{b\gamma}}, \sqrt{\mathfrak{p}_{c\gamma}}$ ; hence there are four factors in which  $\sqrt{\mathfrak{p}_{b\beta}}, \sqrt{\mathfrak{p}_{b\gamma}}$  have determinate signs, but in which we attribute to the radicals  $\sqrt{\mathfrak{p}_{c\beta}}, \sqrt{\mathfrak{p}_{c\gamma}}$  the signs  $+$  or  $-$  at pleasure. It is to be shown that the four factors each vanish for a point on the line  $ab$ ; that is, on writing therein for  $x, y, z, w$  the values  $ux_a + vx_b, uy_a + vy_b, \&c$ .

*p. 120 [source PDF p. 503]*

But we thus have, as before,  $\mathfrak{p}_{a\alpha} = -v \cdot ab\alpha$  and  $\mathfrak{p}_{b\alpha} = u \cdot ab\alpha$ , with the like formulae with  $\beta$  and  $\gamma$  in place of  $\alpha$ . The factor thus becomes

$$\sqrt{-uv} \begin{vmatrix} \sqrt{ab\alpha} & \sqrt{ab\alpha} & \sqrt{\mathfrak{p}_{c\alpha}} \\ \sqrt{ab\beta} & \sqrt{ab\beta} & \sqrt{\mathfrak{p}_{c\beta}} \\ \sqrt{ab\gamma} & \sqrt{ab\gamma} & \sqrt{\mathfrak{p}_{c\gamma}} \end{vmatrix},$$

which vanishes, being  $= O^1$ ; and the norm is thus  $= O^4$ , viz. the line is 4-tuple.

(2) The line  $\alpha$  is 8-tuple: in fact, for a point on the line we have  $\mathfrak{p}_{a\alpha} = 0, \mathfrak{p}_{b\alpha} = 0, \mathfrak{p}_{c\alpha} = 0$ , whence each factor vanishes, being  $= O^{1/2}$ , and the norm is therefore  $O^8$ .

(3) The line  $(ab, \alpha, \beta, \gamma)$  is 4-tuple: in fact, writing  $z = 0, w = 0$  for the equations of the line, we have  $h_\alpha = 0, h_\beta = 0, h_\gamma = 0$ , and  $z_a w_b - z_b w_a = 0$ , or say  $w_a = \lambda z_a, w_b = \lambda z_b$ . Hence, writing

$$I = (g - \lambda a)x - (f + \lambda b)y,$$

viz.  $I_\alpha = (g_\alpha - \lambda a_\alpha)x - (f_\alpha + \lambda b_\alpha)y$ , &c., for  $z = 0$ ,  $w = 0$ , we have  $\mathfrak{p}_{a\alpha} = z_a I_\alpha$ ,  $\mathfrak{p}_{b\alpha} = z_b I_\alpha$ ; and similarly  $\mathfrak{p}_{a\beta} = z_a I_\beta$ ,  $\mathfrak{p}_{b\beta} = z_b I_\beta$ , and  $\mathfrak{p}_{a\gamma} = z_a I_\gamma$ ,  $\mathfrak{p}_{b\gamma} = z_b I_\gamma$ . The factor thus is

$$\sqrt{z_a z_b} \begin{vmatrix} \sqrt{I_\alpha} & \sqrt{I_\alpha} & \sqrt{\mathfrak{p}_{c\alpha}} \\ \sqrt{I_\beta} & \sqrt{I_\beta} & \sqrt{\mathfrak{p}_{c\beta}} \\ \sqrt{I_\gamma} & \sqrt{I_\gamma} & \sqrt{\mathfrak{p}_{c\gamma}} \end{vmatrix},$$

which vanishes, being  $= O^1$ ; there are four such factors, or the norm is  $O^4$ ; whence the line is 4-tuple.

(8) The line  $abc.\alpha.\beta$  is a 4-tuple line. To prove it, take as before  $w = 0$  for the equation of the plane  $abc$ , and  $(z = 0, w = 0)$  for the equations of the line in question. We have  $h_\alpha = 0$ ,  $h_\beta = 0$ ,  $w_a = 0$ ,  $w_b = 0$ ,  $w_c = 0$ ; whence (if  $z = 0$ ,  $w = 0$ ), writing for shortness  $I = gx - fy$  (viz.  $I_\alpha = g_\alpha x - f_\alpha y$ ,  $I_\beta = g_\beta x - f_\beta y$ ), we have

$$\mathfrak{p}_{a\alpha}, \mathfrak{p}_{b\alpha}, \mathfrak{p}_{c\alpha} = I_\alpha z_a, I_\alpha z_b, I_\alpha z_c,$$

and similarly

$$\mathfrak{p}_{a\beta}, \mathfrak{p}_{b\beta}, \mathfrak{p}_{c\beta} = I_\beta z_a, I_\beta z_b, I_\beta z_c.$$

The factor thus is

$$\begin{vmatrix} \sqrt{I_\alpha z_a} & \sqrt{I_\alpha z_b} & \sqrt{I_\alpha z_c} \\ \sqrt{I_\beta z_a} & \sqrt{I_\beta z_b} & \sqrt{I_\beta z_c} \\ \sqrt{\mathfrak{p}_{a\gamma}} & \sqrt{\mathfrak{p}_{b\gamma}} & \sqrt{\mathfrak{p}_{c\gamma}} \end{vmatrix},$$

which vanishes, being  $= O^1$ ; and there are four such factors, obtained by giving to the radicals the signs  $+$ ,  $-$  at pleasure: hence the norm is  $= O^4$ .

*{Surface  $abc\alpha\beta\gamma$ .}*

*p. 121 [source PDF p. 503]*

### Surface $ab\alpha\beta\gamma\delta$ .

35. The equation is

$$\text{Norm}\{\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}}\mathfrak{p}_{\beta\gamma\delta}^2 - \sqrt{\mathfrak{p}_{a\beta}\mathfrak{p}_{b\beta}}\mathfrak{p}_{\gamma\delta\alpha}^2 + \sqrt{\mathfrak{p}_{a\gamma}\mathfrak{p}_{b\gamma}}\mathfrak{p}_{\delta\alpha\beta}^2 - \sqrt{\mathfrak{p}_{a\delta}\mathfrak{p}_{b\delta}}\mathfrak{p}_{\alpha\beta\gamma}^2\} = 0,$$

where the norm is the product of 8 factors each of the order 3. As before,  $\mathfrak{p}_{a\alpha} = 0$  is the equation of the plane through the point  $a$  and the line  $\alpha$ ; viz.  $\mathfrak{p}_{a\alpha}$  has the value previously mentioned; and  $\mathfrak{p}_{\beta\gamma\delta}^2 = 0$  is the equation of the quadric surface through the lines  $\beta, \gamma, \delta$ .

36. *Investigation.* In the projection, taking  $\xi, \eta, \zeta$  as current line-coordinates, the equation of the conic passing through the projections of the points  $a, b$  is

$$\sqrt{(p_a\xi + q_a\eta + r_a\zeta)(p_b\xi + q_b\eta + r_b\zeta)} + A\xi + B\eta + C\zeta = 0,$$

where  $A, B, C$  are arbitrary coefficients. To make this touch the projection of the line  $\alpha$ , we must write  $\xi : \eta : \zeta = P_\alpha : Q_\alpha : R_\alpha$ ; and then

$$\begin{aligned} p_a\xi + q_a\eta + r_a\zeta &= p_aP_\alpha + q_aQ_\alpha + r_aR_\alpha \\ &= w_a(xP_\alpha + yQ_\alpha + zR_\alpha) \\ &\quad - w(x_aP_\alpha + y_aQ_\alpha + z_aR_\alpha) \\ &= -w(x_aP_\alpha + y_aQ_\alpha + z_aR_\alpha + w_aS_\alpha) \\ &= -w\mathfrak{p}_{a\alpha}, \end{aligned}$$

and similarly

$$p_b\xi + q_b\eta + r_b\zeta = -w\mathfrak{p}_{b\alpha}.$$

Hence the equation is

$$w\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}} + AP_\alpha + BQ_\alpha + CR_\alpha = 0;$$

and forming the like equations for the lines  $\beta, \gamma, \delta$  respectively, and eliminating, we have

$$\begin{vmatrix} \sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}} & P_\alpha & Q_\alpha & R_\alpha \\ \sqrt{\mathfrak{p}_{a\beta}\mathfrak{p}_{b\beta}} & P_\beta & Q_\beta & R_\beta \\ \sqrt{\mathfrak{p}_{a\gamma}\mathfrak{p}_{b\gamma}} & P_\gamma & Q_\gamma & R_\gamma \\ \sqrt{\mathfrak{p}_{a\delta}\mathfrak{p}_{b\delta}} & P_\delta & Q_\delta & R_\delta \end{vmatrix} = 0,$$

which, throwing out a factor  $w$ , becomes the displayed equation above; or, taking the norm, we have the above-written equation.

37. *Singularities.* The equation shows that

$$(0) \quad \text{The point } a \text{ is a 4-conical point; for it } \mathfrak{p}_{a\alpha} = \mathfrak{p}_{a\beta} = \mathfrak{p}_{a\gamma} = \mathfrak{p}_{a\delta} = 0,$$

each factor is  $= O^{1/2}$ , and the norm is  $= O^4$ .

*{Surface } ab\alpha\beta\gamma\delta.*

*p. 122 [source PDF p. 503]*

(1) The line  $ab$  is a 2-tuple line. To prove this, we have for the coordinates of a point on the line in question  $ux_a + vx_b$ ,  $uy_a + vy_b$ , &c.; the values of  $\mathfrak{p}_{a\alpha}$ ,  $\mathfrak{p}_{b\alpha}$  become as before  $-v \cdot ab\alpha$ ,  $+u \cdot ab\alpha$ , and similarly for  $\mathfrak{p}_{a\beta}$ ,  $\mathfrak{p}_{b\beta}$ , &c.; so that, omitting the constant factor  $\sqrt{-uv}$ , the value of the factor is

$$ab\alpha \cdot \mathfrak{p}_{\beta\gamma\delta}^2 - ab\beta \cdot \mathfrak{p}_{\gamma\delta\alpha}^2 + ab\gamma \cdot \mathfrak{p}_{\delta\alpha\beta}^2 - ab\delta \cdot \mathfrak{p}_{\alpha\beta\gamma}^2.$$

Taking  $(a, b, c, f, g, h)$  for the coordinates of the line  $ab$ , we have

$$ab\alpha = af_\alpha + bg_\alpha + ch_\alpha + fa_\alpha + gb_\alpha + hc_\alpha,$$

with the like expressions for  $ab\beta$ , &c.; and substituting for  $\mathfrak{p}_{\beta\gamma\delta}^2$ , &c., their values, the factor is represented by the following coefficient table:

|     | $x^2$  | $y^2$  | $z^2$  | $w^2$ | $xw$          | $yw$          | $zw$          | $yz$          | $zx$          | $xy$          |
|-----|--------|--------|--------|-------|---------------|---------------|---------------|---------------|---------------|---------------|
| $a$ | $fagh$ |        |        | $fab$ | $fabg - fach$ | $fbch$        | $-fbcg$       |               | $fcgh$        | $fbgh$        |
| $b$ |        | $gbhf$ |        | $gab$ | $-gach$       | $gbch - gabf$ | $gcaf$        | $gchf$        | $hafg$        | $gahf$        |
| $c$ |        |        | $hcfg$ | $hac$ | $habg$        | $-habf$       | $hcaf - hbeg$ | $hbfg$        |               |               |
| $f$ |        | $abhf$ | $acfg$ |       | .             | $abch$        | $-abeg$       | $abfg + achf$ | $acgh$        | $abgh$        |
| $g$ | $bagh$ |        | $bcfg$ |       | $-bach$       |               | $bcaf$        | $bchf$        | $bcgh + bafg$ | $bahf$        |
| $h$ | $cagh$ | $cbhf$ |        |       | $cabg$        | $-cabf$       |               | $cbfg$        | $caf$         | $cahf + cbgh$ |

Viz. the value of the factor is  $\{a(fagh) + g(bagh) + h(cagh)\}x^2 + \&c.$ , where  $fagh = f_{\alpha}a_{\beta}g_{\gamma}h_{\delta}$  is the determinant

$$\begin{vmatrix} f & a & g & h \\ \vdots & \vdots & \vdots & \vdots \end{vmatrix},$$

the suffixes in the four lines being  $\alpha, \beta, \gamma, \delta$  respectively. Collecting, this is

$$\begin{aligned} & (\quad + cbhfy - bcfgz + abcw)(\quad + hy - gz + aw) \\ & + (-caghx \quad + acfgz + abcw)(-hx \quad + fz + bw) \\ & + (baghx - abhfy \quad + habcw)(gx - fy \quad + cw) \\ & + (-afghx - bfg hy - cfghz \quad)(ax + by + cz \quad) \\ & + bcgh\{w(ax + by + cz) - x(\quad hy - gz + aw)\} \\ & + cahf\{w(ax + by + cz) - y(-hx + fz + bw)\} \\ & + abfg\{w(ax + by + cz) - z(gx - fy + cw)\} = 0; \end{aligned}$$

or, what is the same thing,

$$AP + BQ + CR + DS = 0.$$

*{Surface abcd $\alpha\beta$ .}*

*p. 123 [source PDF p. 503]*

where

$$\begin{aligned} A &= \{-bcghx + cbhfy - bcfgz + abcw\}, \\ B &= \{-caghx + cahfy + acfgz + abcw\}, \\ C &= \{ \quad baghx - abhfy + abfgz + habcw\}, \\ D &= \{-afghx - bfg hy - cfghz + (bcgh + cahf + abfg)w\}, \\ P &= (\quad + hy - gz + aw), \\ Q &= (-hx \quad + fz + bw), \\ R &= (gx - fy \quad + cw), \\ S &= (ax + by + cz \quad) = 0, \end{aligned}$$

the right-hand factors vanishing for the values  $ux_a + vx_b$  of the coordinates.

38. It thus appears so far that the factor is  $= O^1$ ; it is, in fact,  $= O^2$ , viz. we can show that, operating upon it with

$$\Delta = X\delta_x + Y\delta_y + Z\delta_z + W\delta_w,$$

the value (for any point of the line  $ab$ ) is  $= 0$ . We have

$$\Delta\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}}\mathfrak{p}_{\beta\gamma\delta}^2 = \frac{\mathfrak{p}_{a\alpha}l_{b\alpha} + \mathfrak{p}_{b\alpha}l_{a\alpha}}{2\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}}}\mathfrak{p}_{\beta\gamma\delta}^2 + \sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}}\Delta\mathfrak{p}_{\beta\gamma\delta}^2,$$

where  $l_{b\alpha}(= \Delta\mathfrak{p}_{b\alpha})$  is what  $\mathfrak{p}_{b\alpha}$  becomes on writing therein  $(X, Y, Z, W)$  in place of  $(x, y, z, w)$ . Writing, as before, for  $x, y, z, w$  the values  $ux_a + vx_b$ , &c., we have

$$\mathfrak{p}_{a\alpha} = -v \cdot ab\alpha, \quad \mathfrak{p}_{b\alpha} = u \cdot ab\alpha;$$

and putting for shortness

$$-v \cdot l_{b\alpha} + u \cdot l_{a\alpha} = lk\alpha, \quad \&c.,$$

the expression in question, divided by  $\sqrt{-uv}$ , is

$$= -2vu\{ab\alpha \cdot \Delta\mathfrak{p}_{\beta\gamma\delta}^2 - \&c.\} + \{lk\alpha \cdot \mathfrak{p}_{\beta\gamma\delta}^2 - \&c.\},$$

where, denoting the determinants

$$\begin{vmatrix} X & Y & Z & W \\ ux_a - vx_b & uy_a - vy_b & uz_a - vz_b & uw_a - vw_b \end{vmatrix}$$

by  $(a', b', c', f', g', h')$ , we have

$$lk\alpha = a'f_\alpha + b'g_\alpha + c'h_\alpha + f'a_\alpha + g'b_\alpha + h'c_\alpha.$$

But

$$ab\alpha \cdot \Delta \mathfrak{p}_{\beta\gamma\delta}^2 = \Delta ab\alpha \cdot \mathfrak{p}_{\beta\gamma\delta}^2,$$

since  $ab\alpha$  is independent of  $(x, y, z, w)$ ; and the expression is

$$= -2vu\Delta(AP + BQ + CR + DS) + AP' + BQ' + CR' + DS'.$$

*p. 124 [source PDF p. 503]*

Here  $P', Q', R', S'$  denote  $h'y - g'z + a'w$ , &c.; and where, finally,  $x, y, z, w$  are to be replaced by  $ux_a + vx_b$ , &c. Since for these values  $P, Q, R, S$  vanish, the expression becomes

$$= -2vu(A\Delta P + B\Delta Q + C\Delta R + D\Delta S) + AP' + BQ' + CR' + DS',$$

that is

$$= A(P' - 2vu\Delta P) + B(Q' - 2vu\Delta Q) + C(R' - 2vu\Delta R) + D(S' - 2vu\Delta S).$$

And we have, in fact,  $P' - 2vu\Delta P = 0$ , &c. For, writing for a moment

$$\begin{aligned} x, y, z, w &= ux_a + vx_b, \quad uy_a + vy_b, \quad uz_a + vz_b, \quad uw_a + vw_b, \\ x', y', z', w' &= ux_a - vx_b, \quad uy_a - vy_b, \quad uz_a - vz_b, \quad uw_a - vw_b, \end{aligned}$$

then, for instance,

$$S' = a'x + b'y + c'z,$$

where

$$a', b', c' = Yz' - Zy', \quad Zx' - Xz', \quad Xy' - Yx';$$

and thence

$$S' = - \begin{vmatrix} X & Y & Z \\ x & y & z \\ x' & y' & z' \end{vmatrix} = 2uv(aX + bY + cZ) = 2uv\Delta S;$$

and similarly for the other equations. The factor is thus  $= O^2$ ; there is only one such factor, and the line  $ab$  is double.

(2) The line  $\alpha$  is an 8-tuple line: in fact, for a point on the line we have  $\mathfrak{p}_{a\alpha} = 0$ ,  $\mathfrak{p}_{b\alpha} = 0$ ,  $\mathfrak{p}_{\gamma\delta\alpha}^2 = 0$ ,  $\mathfrak{p}_{\delta\alpha\beta}^2 = 0$ ,  $\mathfrak{p}_{\alpha\beta\gamma}^2 = 0$ ; and the factor vanishes, being  $= O^1$ . Each of the factors is  $O^1$ , and the norm is  $= O^8$ .

39. (3) The line  $[ab, \alpha, \beta, \gamma]$  is a double line. To prove this, observe first that for a point on this line we have  $\mathfrak{p}_{\alpha\beta\gamma}^2 = 0$ .

Taking as before  $z = 0$ ,  $w = 0$  for the equation of the line  $ab, \alpha, \beta, \gamma$ , we have  $h_\alpha = 0$ ,  $h_\beta = 0$ ,  $h_\gamma = 0$ , and  $z_a w_b - z_b w_a = 0$ ; or say  $w_a = \lambda z_a$ ,  $w_b = \lambda z_b$ ; whence, writing for shortness

$$I = -(g - \lambda a)x + (f + \lambda b)y,$$

viz.  $I_\alpha = -(g_\alpha - \lambda a_\alpha)x + (f_\alpha + \lambda b_\alpha)y$ , we have (when  $z = 0$ ,  $w = 0$ )

$$\mathfrak{p}_{a\alpha} = z_a I_\alpha, \quad \mathfrak{p}_{b\alpha} = z_b I_\alpha,$$



or, omitting the factor  $\sqrt{z_a z_b}$ ,  $\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}} = I_\alpha$ ; and so for  $\sqrt{\mathfrak{p}_{a\beta}\mathfrak{p}_{b\beta}}$  and  $\sqrt{\mathfrak{p}_{a\gamma}\mathfrak{p}_{b\gamma}}$ . The factor thus is

$$I_\alpha \mathfrak{p}_{\beta\gamma\delta}^2 - I_\beta \mathfrak{p}_{\gamma\delta\alpha}^2 + I_\gamma \mathfrak{p}_{\delta\alpha\beta}^2;$$

viz. writing  $z = 0$ ,  $w = 0$  in the expressions of  $\mathfrak{p}_{\beta\gamma\delta}^2$ , &c., this may be written

$$\Sigma[(g - \lambda a)x - (f + \lambda b)y]\{(agh)x^2 + [(ahf) + (bgh)]xy + (bhf)y^2\},$$

where observe that  $\Sigma$  denotes a sum of three terms of the form

$$\alpha \cdot \beta\gamma\delta - \beta \cdot \gamma\delta\alpha + \gamma \cdot \delta\alpha\beta.$$

*{Surface  $abcd\alpha\beta$ .}*

*p. 125 [source PDF p. 503]*

Adding thereto a fourth term  $-\delta \cdot \alpha\beta\gamma$ , the value of the sum would be  $= \alpha\beta\gamma\delta$ , or the sum of the three terms is  $= \alpha\beta\gamma\delta + \delta \cdot \alpha\beta\gamma$ , where the symbols represent determinants. But in each case the determinant  $\alpha\beta\gamma$  is  $= 0$ , as containing the column  $h_\alpha, h_\beta, h_\gamma$ , the terms of which are each  $= 0$ : thus  $\Sigma g \cdot agh$  is  $gagh - g_\delta \cdot agh$ , where in  $gagh$  the suffixes are  $\alpha, \beta, \gamma, \delta$ , and in  $agh$  they are  $\alpha, \beta, \gamma$ : that is, we have  $\Sigma g \cdot agh = gagh$ . And the whole expression thus is

$$\begin{aligned} &= x^3(gagh - \lambda aagh) \\ &\quad + x^2y(gahf - \lambda aahf + gbgh - \lambda abgh - fagh - \lambda bagh) \\ &\quad + xy^2(gbhf - \lambda abhf - fahf - \lambda bahf - fbgh - \lambda bbgh) \\ &\quad + y^3(-fbbf - \lambda bbbf), \end{aligned}$$

where  $gahf$  denotes the determinant

$$\begin{vmatrix} g & a & h & f \\ \vdots & \vdots & \vdots & \vdots \end{vmatrix}$$

with the suffixes  $\alpha, \beta, \gamma, \delta$  in the four lines respectively, and so in other cases. The terms, such as  $gagh$ , which contain a twice-repeated letter, vanish of themselves; and in the coefficients of  $x^2y$  and  $xy^2$ , the terms which do not separately vanish destroy each other in pairs,  $gahf - fagh = 0$ , &c.; whence the factor vanishes, being  $= O^1$ . There are two such factors (viz. the zero term  $\sqrt{\mathfrak{p}_{a\delta}\mathfrak{p}_{b\delta}}\mathfrak{p}_{\alpha\beta\gamma}^2$  may be taken with the sign  $+$  or  $-$  at pleasure), and the norm is thus  $= O^2$ .

40. But the line is tacnodal, each sheet of the surface touching along the line in question the hyperboloid  $\mathfrak{p}_{\alpha\beta\gamma}^2$ . To prove this, write

$$\Delta = X\delta_x + Y\delta_y + Z\delta_z + W\delta_w;$$

we have for the hyperboloid, writing  $z = 0$ ,  $w = 0$ ,

$$\Delta \mathfrak{p}_{\alpha\beta\gamma}^2 = [(afg \cdot x + bfg \cdot y)Z + (abg \cdot x - abf \cdot y)W];$$

and it is to be shown that

$$\Delta(\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}}\mathfrak{p}_{\beta\gamma\delta}^2 - \sqrt{\mathfrak{p}_{a\beta}\mathfrak{p}_{b\beta}}\mathfrak{p}_{\gamma\delta\alpha}^2 + \sqrt{\mathfrak{p}_{a\gamma}\mathfrak{p}_{b\gamma}}\mathfrak{p}_{\delta\alpha\beta}^2 \mp \sqrt{\mathfrak{p}_{a\delta}\mathfrak{p}_{b\delta}}\mathfrak{p}_{\alpha\beta\gamma}^2)$$

each contain the factor  $\Delta \mathfrak{p}_{\alpha\beta\gamma}^2$ ; or, what is the same thing, that

$$\Delta \Sigma \sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}}\mathfrak{p}_{\beta\gamma\delta}^2$$

contains the factor in question,  $\Sigma$  denoting the sum of the first three terms of the original expression. The value is

$$= \Sigma \left\{ \frac{\mathfrak{p}_{a\alpha}P_{b\alpha} + \mathfrak{p}_{b\alpha}P_{a\alpha}}{2\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}}} \mathfrak{p}_{\beta\gamma\delta}^2 + \sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}} \Delta \mathfrak{p}_{\beta\gamma\delta}^2 \right\};$$

where  $P_{a\alpha} = \Delta \mathbf{p}_{a\alpha}$  denotes what  $\mathbf{p}_{a\alpha}$  becomes on writing therein  $X, Y, Z, W$  for  $x, y, z, w$ ; and the like as to  $P_{b\alpha}$ . Substituting for  $\mathbf{p}_{a\alpha}$  and  $\mathbf{p}_{b\alpha}$  their values  $z_a I_\alpha$  and  $z_b I_\alpha$ , and multiplying by  $\sqrt{z_a z_b}$ , the expression is

$$= \Sigma \{ (z_a P_{b\alpha} + z_b P_{a\alpha}) \mathbf{p}_{\beta\gamma\delta}^2 + 2z_a z_b I_\alpha \Delta \mathbf{p}_{\beta\gamma\delta}^2 \}.$$

{Surface  $abcd\alpha\beta$ .}

p. 126 [source PDF p. 503]

where we have

$$\begin{aligned} z_a P_{b\alpha} + z_b P_{a\alpha} &= z_b \{ x_a (Zg_\alpha - Wa_\alpha) + y_a (-Zf_\alpha - Wb_\alpha) \\ &\quad + z_a [X(-g_\alpha + \lambda a_\alpha) + Y(f_\alpha + \lambda b_\alpha) + (\lambda Z - W)c_\alpha] \} \\ &\quad + z_a \{ x_b (Zg_\alpha - Wa_\alpha) + y_b (-Zf_\alpha - Wb_\alpha) \\ &\quad + z_b [X(-g_\alpha + \lambda a_\alpha) + Y(f_\alpha + \lambda b_\alpha) + (\lambda Z - W)c_\alpha] \} \\ &= (z_b x_a + z_a x_b)(Zg_\alpha - Wa_\alpha) \\ &\quad + (z_b y_a + z_a y_b)(-Zf_\alpha - Wb_\alpha) \\ &\quad + 2z_a z_b \{ X(-g_\alpha + \lambda a_\alpha) + Y(f_\alpha + \lambda b_\alpha) + (\lambda Z - W)c_\alpha \}. \end{aligned}$$

Also

$$\begin{aligned} z_a z_b I_\alpha &= z_a z_b \{ (-g_\alpha + \lambda a_\alpha)x + (f_\alpha + \lambda b_\alpha)y \}, \\ \mathbf{p}_{\beta\gamma\delta}^2 &= x^2 \cdot agh + xy(ahf + bgh) + y^2 \cdot bhf, \\ \Delta \mathbf{p}_{\beta\gamma\delta}^2 &= X \{ 2x \cdot agh + y(ahf + bgh) \} \\ &\quad + Y \{ x(ahf + bgh) + 2y \cdot bhf \} \\ &\quad + Z \{ x(cgh + afg) + y(bfg + chf) \} \\ &\quad + W \{ x(abg - cah) + y(bch - abf) \}. \end{aligned}$$

41. The whole expression is a linear function of  $X, Y, Z, W$ , and it is easy to see *a priori*, or to verify, that the coefficients of  $X, Y$ , each of them vanish. The coefficient of  $Z$  is

$$\begin{aligned} &= \Sigma \{ (z_b x_a + z_a x_b)g_\alpha - (z_b y_a + z_a y_b)f_\alpha + 2\lambda z_a z_b c_\alpha \} \mathbf{p}_{\beta\gamma\delta}^2 \\ &\quad + \Sigma 2z_a z_b [(-g_\alpha + \lambda a_\alpha)x + (f_\alpha + \lambda b_\alpha)y] [x(cgh + afg) + y(bfg + chf)], \end{aligned}$$

with a like expression for the coefficient of  $W$ .

The foregoing expression may be written

$$\begin{aligned} &(z_b x_a + z_a x_b) \Sigma g [agh \cdot x^2 + (ahf + bgh)xy + bhf \cdot y^2] \\ &\quad - (z_b y_a + z_a y_b) \Sigma f [agh \cdot x^2 + (ahf + bgh)xy + bhf \cdot y^2] \\ &\quad + 2\lambda z_a z_b \Sigma \{ c [agh \cdot x^2 + (ahf + bgh)xy + bhf \cdot y^2] \\ &\quad \quad + (ax + by) [(cgh + afg)x + (bfg + chf)y] \} \\ &\quad + 2z_a z_b \Sigma (-gx + fy) [(cgh + afg)x + (bfg + chf)y]. \end{aligned}$$

The first sum is

$$x^2 \cdot gagh + xy(gahf + gbgh) + y^2 \cdot gbhf = -xy \cdot afg h - y^2 \cdot bfg h = -h_\delta y (afg \cdot x + bfg \cdot y);$$

where  $afg, bfg$  denote determinants with the suffixes  $\alpha, \beta, \gamma$ . Similarly the second sum is

$$= -h_\delta x (afg \cdot x + bfg \cdot y);$$

the third sum is

$$(a_\delta x + b_\delta y) (afg \cdot x + bfg \cdot y),$$

and the fourth sum is

$$(-g_\delta x + f_\delta y) (afg \cdot x + bfg \cdot y).$$

*{Surface  $abcd\alpha\beta$ .}*

*p. 127 [source PDF p. 503]*

The whole coefficient of  $Z$  thus contains the factor  $(afg \cdot x + bfg \cdot y)$ ; and similarly it would appear that the whole coefficient of  $W$  contains the factor  $(abg \cdot x - abf \cdot y)$ , the other factor being the same in each case; viz. the two terms together are

$$\left\{ \begin{array}{l} -(z_b x_a + z_a x_b) h_\delta y \\ +(z_b y_a + z_a y_b) h_\delta x \\ +2\lambda z_a z_b (a_\delta x + b_\delta y) \\ +2z_a z_b (-g_\delta x + f_\delta y) \end{array} \right\} \{Z(afg \cdot x + bfg \cdot y) + W(abg \cdot x - abf \cdot y)\};$$

where the second factor is  $\Delta p_{\alpha\beta\gamma}^2$ , which is the required result. See post, Nos. 59 et seq.

42. (4) The line  $[\alpha, \beta, \gamma, \delta]$  is an 8-tuple line; in fact, for any point of the line in question we have

$$p_{\beta\gamma\delta}^2 = 0, \quad p_{\gamma\delta\alpha}^2 = 0, \quad p_{\delta\alpha\beta}^2 = 0, \quad p_{\alpha\beta\gamma}^2 = 0;$$

whence each factor is  $O^1$ , or the norm is  $O^8$ .

I notice that the surface meets the quadric  $p_{\alpha\beta\gamma}^2$  in

|                                        |              |                     |
|----------------------------------------|--------------|---------------------|
| lines $\alpha, \beta, \gamma$          | each 8 times | 24,                 |
| line $(\alpha, \beta, \gamma, \delta)$ |              | 16,                 |
| lines $(ab, \alpha, \beta, \gamma)$    | each 4 times | 8,                  |
|                                        |              | $24 \times 2 = 48.$ |

**Surface  $a\alpha\beta\gamma\delta\epsilon$ .**

43. The equation is

$$\begin{aligned} & (p_{\alpha\beta\epsilon}^2 p_{\gamma\delta\epsilon}^2 p_{a\alpha\gamma}^3 \cdot \delta\beta + p_{\alpha\gamma\epsilon}^2 p_{\delta\beta\epsilon}^2 p_{a\alpha\beta}^3 \cdot \gamma\delta)^2 \\ & - 4p_{\alpha\beta\epsilon}^2 p_{\gamma\delta\epsilon}^2 p_{\alpha\gamma\epsilon}^2 p_{\delta\beta\epsilon}^2 p_{\alpha\beta\gamma}^2 p_{a\alpha} p_{\delta a} = 0; \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} & (p_{\alpha\beta\epsilon}^2 p_{\gamma\delta\epsilon}^2 p_{a\alpha\gamma}^3 \cdot \delta\beta - p_{\alpha\gamma\epsilon}^2 p_{\delta\beta\epsilon}^2 p_{a\alpha\beta}^3 \cdot \gamma\delta)^2 \\ & - 4p_{\alpha\beta\epsilon}^2 p_{\gamma\delta\epsilon}^2 p_{\alpha\gamma\epsilon}^2 p_{\delta\beta\epsilon}^2 p_{\beta\alpha\delta}^2 p_{\gamma\alpha\delta}^2 p_{\beta a} p_{\gamma a} = 0; \end{aligned}$$

the equivalence of the two depending on the identity

$$p_{\alpha\beta}^3 \cdot \gamma\delta \cdot p_{a\alpha\gamma}^3 \cdot \delta\beta - p_{\alpha\beta\gamma}^2 p_{\delta\beta\gamma}^2 p_{a\alpha} p_{\delta a} + p_{\beta\alpha\delta}^2 p_{\gamma\alpha\delta}^2 p_{\beta a} p_{\gamma a} = 0;$$

where, as before,  $p_{\alpha\beta\epsilon}^2 = 0$  is the equation of the quadric through the lines  $\alpha, \beta, \epsilon$ , and  $p_{a\alpha} = 0$  is the equation of the plane through the line  $\alpha$  and the point  $a$ ; viz.  $p_{\alpha\beta\epsilon}^2$ , &c., and  $p_{a\alpha}$ , &c., have the values already mentioned:  $p_{a\alpha\beta}^3 \cdot \gamma\delta = 0$  as already mentioned is the cubic surface through the lines  $\alpha, \beta, \gamma, \delta$  and  $a\alpha\beta, a\gamma\delta$ .

*{Surface  $a\alpha\beta\gamma\delta\epsilon$ .}*

*p. 128 [source PDF p. 503]*

44. *Investigation.* In the projection, using line-coordinates, the equation of the conic touching the five lines may be written

$$\left| \begin{array}{c} (\xi, \eta, \zeta)^2 \\ (P, Q, R)^2 \end{array} \right| = 0;$$

where the symbol denotes a determinant the last five lines of which are obtained by giving to  $(P, Q, R)$  the suffixes  $\alpha, \beta, \gamma, \delta, \epsilon$  respectively. This is at once transformed into

$$\alpha\beta\epsilon \cdot \gamma\delta\epsilon \cdot \alpha\gamma\Delta \cdot \delta\beta\Delta - \alpha\gamma\epsilon \cdot \delta\beta\epsilon \cdot \alpha\beta\Delta \cdot \gamma\delta\Delta = 0,$$

or, what is the same thing,

$$\mathfrak{p}_{\alpha\beta\epsilon}^2 \mathfrak{p}_{\gamma\delta\epsilon}^2 \alpha\gamma\Delta \cdot \delta\beta\Delta - \mathfrak{p}_{\alpha\gamma\epsilon}^2 \mathfrak{p}_{\delta\beta\epsilon}^2 \alpha\beta\Delta \cdot \gamma\delta\Delta = 0;$$

or say

$$\begin{aligned} & \mathfrak{p}_{\alpha\beta\epsilon}^2 \mathfrak{p}_{\gamma\delta\epsilon}^2 (A''\xi + B''\eta + C''\zeta)(A'''\xi + B'''\eta + C'''\zeta) \\ & - \mathfrak{p}_{\alpha\gamma\epsilon}^2 \mathfrak{p}_{\delta\beta\epsilon}^2 (A\xi + B\eta + C\zeta)(A'\xi + B'\eta + C'\zeta) = 0; \end{aligned}$$

where  $\mathfrak{p}_{\alpha\beta\epsilon}^2$ , &c., signify as before; and

$$A\xi + B\eta + C\zeta = \begin{vmatrix} \xi & \eta & \zeta \\ P_\alpha & Q_\alpha & R_\alpha \\ P_\beta & Q_\beta & R_\beta \end{vmatrix},$$

and so for  $A'\xi + B'\eta + C'\zeta$ , &c., the suffixes for  $A', B', C'$  being  $(\gamma, \delta)$ ; and those for  $A''\xi + B''\eta + C''\zeta$  and  $A'''\xi + B'''\eta + C'''\zeta$  being  $(\alpha, \gamma)$  and  $(\delta, \beta)$  respectively.

45. Passing to the reciprocal equation, and making the conic pass through the point  $a$ , we obtain the equation of the surface in the form

$$\begin{aligned} & \left\{ \mathfrak{p}_{\alpha\gamma\epsilon}^2 \mathfrak{p}_{\delta\beta\epsilon}^2 \begin{vmatrix} p_a & q_a & r_a \\ A & B & C \\ A' & B' & C' \end{vmatrix} - \mathfrak{p}_{\alpha\beta\epsilon}^2 \mathfrak{p}_{\gamma\delta\epsilon}^2 \begin{vmatrix} p_a & q_a & r_a \\ A'' & B'' & C'' \\ A''' & B''' & C''' \end{vmatrix} \right\}^2 \\ & + 4\mathfrak{p}_{\alpha\gamma\epsilon}^2 \mathfrak{p}_{\delta\beta\epsilon}^2 \mathfrak{p}_{\alpha\beta\epsilon}^2 \mathfrak{p}_{\gamma\delta\epsilon}^2 \begin{vmatrix} p_a & q_a & r_a \\ A & B & C \\ A'' & B'' & C'' \end{vmatrix} \begin{vmatrix} p_a & q_a & r_a \\ A' & B' & C' \\ A''' & B''' & C''' \end{vmatrix} = 0; \end{aligned}$$

or in the equivalent form, where in the first term we have  $+$  instead of  $-$ , and in the second term the determinants are

$$\begin{vmatrix} p_a & q_a & r_a \\ A & B & C \\ A''' & B''' & C''' \end{vmatrix}, \quad \begin{vmatrix} p_a & q_a & r_a \\ A' & B' & C' \\ A'' & B'' & C'' \end{vmatrix}.$$

46. To reduce this result, observe that we have

$$A, B, C = \begin{vmatrix} hy - gz + aw & -hx + fz + bw & gx - fy + cw \\ h'y - g'z + a'w & -h'x + f'z + b'w & g'x - f'y + c'w \end{vmatrix}.$$

*p. 129 [source PDF p. 503]*

Here, for convenience, I retain the unaccented and accented letters  $(a, \dots)$ ,  $(a', \dots)$  instead of these letters with the suffixes  $\alpha$  and  $\beta$  respectively. Writing as before

$$\begin{aligned} L &= (af' - a'f)x + \dots, \\ M &= (ag' - a'g)x + \dots, \\ N &= (ah' - a'h)x + \dots, \\ \Omega &= (gh' - g'h)x + \dots, \end{aligned}$$

then

$$A = \Omega x - Lw, \quad B = \Omega y - Mw, \quad C = \Omega z - Nw,$$

and similarly

$$A' = \Omega'x - L'w, \quad B' = \Omega'y - M'w, \quad C' = \Omega'z - N'w;$$

where for  $L', M', N', \Omega'$  we have  $(a'', \dots)$  and  $(a''', \dots)$ . Hence

$$BC' - B'C = w \begin{vmatrix} y & z & w \\ M & N & \Omega \\ M' & N' & \Omega' \end{vmatrix},$$

with like expressions for  $CA' - C'A$  and  $AB' - A'B$ ; and substituting, we have

$$\begin{vmatrix} p_a & q_a & r_a \\ A & B & C \\ A' & B' & C' \end{vmatrix} = w \begin{vmatrix} p_a & q_a & r_a \\ x & y & z & w \\ L & M & N & \Omega \\ L' & M' & N' & \Omega' \end{vmatrix}.$$

Or substituting for  $p_a, q_a, r_a$  their values  $xw_a - wx_a, yw_a - wy_a, zw_a - wz_a$ , this is

$$\begin{vmatrix} p_a & q_a & r_a \\ A & B & C \\ A' & B' & C' \end{vmatrix} = -w^2 \begin{vmatrix} x & y & z & w \\ x_a & y_a & z_a & w_a \\ L & M & N & \Omega \\ L' & M' & N' & \Omega' \end{vmatrix}.$$

Whence, omitting the factors  $w^2$ , the equation is

$$\left\{ \mathfrak{p}_{\alpha\gamma\epsilon}^2 \mathfrak{p}_{\delta\beta\epsilon}^2 \begin{vmatrix} x & y & z & w \\ x_a & y_a & z_a & w_a \\ L & M & N & \Omega \\ L' & M' & N' & \Omega' \end{vmatrix} - \mathfrak{p}_{\alpha\beta\epsilon}^2 \mathfrak{p}_{\gamma\delta\epsilon}^2 \begin{vmatrix} x & y & z & w \\ x_a & y_a & z_a & w_a \\ L'' & M'' & N'' & \Omega'' \\ L''' & M''' & N''' & \Omega''' \end{vmatrix} \right\}^2 \\ + 4\mathfrak{p}_{\alpha\gamma\epsilon}^2 \mathfrak{p}_{\delta\beta\epsilon}^2 \mathfrak{p}_{\alpha\beta\epsilon}^2 \mathfrak{p}_{\gamma\delta\epsilon}^2 \begin{vmatrix} x & y & z & w \\ x_a & y_a & z_a & w_a \\ L & M & N & \Omega \\ L'' & M'' & N'' & \Omega'' \end{vmatrix} \begin{vmatrix} x & y & z & w \\ x_a & y_a & z_a & w_a \\ L' & M' & N' & \Omega' \\ L''' & M''' & N''' & \Omega''' \end{vmatrix} = 0.$$

*p. 130 [source PDF p. 503]*

Here I recall that for  $(L, \dots)$ ,  $(L', \dots)$ ,  $(L'', \dots)$ ,  $(L''', \dots)$  the suffixes are  $(\alpha, \beta)$ ,  $(\gamma, \delta)$ ,  $(\alpha, \gamma)$ , and  $(\delta, \beta)$  respectively. The values of the first two determinants thus are  $\mathfrak{p}_{\alpha\alpha\beta}^3 \cdot \gamma\delta$  and  $\mathfrak{p}_{\alpha\alpha\gamma}^3 \cdot \delta\beta$  respectively: that of the third is  $\mathfrak{p}_{\alpha\alpha\beta}^3 \cdot \alpha\gamma$ ; viz. this is  $= \mathfrak{p}_{\alpha\beta\gamma}^2 \mathfrak{p}_{a\alpha}$ ; similarly, that of the fourth is  $\mathfrak{p}_{\alpha\gamma\delta}^3 \cdot \delta\beta$ , which is  $= -\mathfrak{p}_{a\delta\gamma}^3 \cdot \delta\beta = +\mathfrak{p}_{a\delta\beta}^3 \cdot \delta\gamma$ ; or finally this is  $= \mathfrak{p}_{\delta\beta\gamma}^2 \mathfrak{p}_{a\delta}$ . And we have thus the before-mentioned equation of the surface.

47. *Singularities.* The equation of the surface shows that:

(0) The point  $a$  is a 2-conical point: for it  $\mathfrak{p}_{\alpha\alpha\beta}^3 \cdot \gamma\delta = 0$ ,  $\mathfrak{p}_{\alpha\alpha\gamma}^3 \cdot \delta\beta = 0$ ,  $\mathfrak{p}_{a\alpha} = 0$ ,  $\mathfrak{p}_{a\delta} = 0$ .

(2) The line  $\alpha$  is a 4-tuple line: for any point on it  $\mathfrak{p}_{\alpha\beta\epsilon}^2 = 0$ ,  $\mathfrak{p}_{\alpha\alpha\beta}^3 \cdot \gamma\delta = 0$ ,  $\mathfrak{p}_{\alpha\gamma\epsilon}^2 = 0$ ,  $\mathfrak{p}_{\alpha\alpha\gamma}^3 \cdot \delta\beta = 0$ ,  $\mathfrak{p}_{\alpha\beta\gamma}^2 = 0$ ,  $\mathfrak{p}_{\alpha a}^2 = 0$ .

(4) The line  $(\alpha, \beta, \gamma, \epsilon)$  is a 2-tuple line: for any point on it  $\mathfrak{p}_{\alpha\beta\epsilon}^2 = 0$ ,  $\mathfrak{p}_{\alpha\gamma\epsilon}^2 = 0$ .

(10) The excuboquartic  $\alpha\beta\epsilon.\gamma\delta.a$  is a simple curve. In fact, for any point of this curve we have  $\mathfrak{p}_{\alpha\beta\epsilon}^2 = 0$ ,  $\mathfrak{p}_{\alpha\alpha\beta}^3 \cdot \gamma\delta = 0$ , these two surfaces intersecting in the lines  $\alpha, \beta$  and the curve. It is, moreover, obvious that the surface is touched along the curve by the hyperboloid  $\mathfrak{p}_{\alpha\beta\epsilon}^2$ .

I notice that the surface meets the quadric  $\mathfrak{p}_{\alpha\beta\gamma}^2$  in

|                                                |               |                     |
|------------------------------------------------|---------------|---------------------|
| lines $(\alpha, \beta, \gamma)$                | each 4 times, | 12,                 |
| line $(\alpha, \beta, \gamma, \delta)$         | twice,        | 4,                  |
| line $(\alpha, \beta, \gamma, \epsilon)$       | twice,        | 4,                  |
| curve $\alpha\alpha\beta\gamma.\delta\epsilon$ | twice,        | 8,                  |
|                                                |               | $14 \times 2 = 28.$ |

**Surface  $\alpha\beta\gamma\delta\epsilon\zeta$ .**

48. The equation of the surface may be written

$$\mathfrak{p}_{\alpha\beta\epsilon}^2 \mathfrak{p}_{\gamma\delta\epsilon}^2 \mathfrak{p}_{\alpha\gamma\zeta}^2 \mathfrak{p}_{\delta\beta\zeta}^2 - \mathfrak{p}_{\alpha\beta\zeta}^2 \mathfrak{p}_{\gamma\delta\zeta}^2 \mathfrak{p}_{\alpha\gamma\epsilon}^2 \mathfrak{p}_{\delta\beta\epsilon}^2 = 0,$$

where  $\mathfrak{p}_{\alpha\beta\epsilon}^2 = 0$  is the equation of the quadric through the lines  $\alpha, \beta, \epsilon$ ; viz.  $\mathfrak{p}_{\alpha\beta\epsilon}^2$  has the value already mentioned.

The form is one of 45 like forms depending on the partitionment

$$\left\{ \begin{array}{l} \alpha\beta.\gamma\delta \\ \alpha\gamma.\delta\beta \\ \alpha\delta.\beta\gamma \end{array} \right\} (\epsilon, \zeta)$$

of the six letters.

{Surface  $\alpha\beta\gamma\delta\epsilon\zeta$ .}

p. 131 [source PDF p. 503]

49. *Investigation.* The projections of the six lines are tangents to a conic: the condition for this is  $(P, Q, R)^2 = 0$ , where the left-hand side represents the determinant obtained by writing successively  $(P_\alpha, Q_\alpha, R_\alpha)$ , &c., for  $(P, Q, R)$ . The equation may be written

$$\alpha\beta\epsilon \cdot \gamma\delta\epsilon \cdot \alpha\gamma\zeta \cdot \delta\beta\zeta - \alpha\beta\zeta \cdot \gamma\delta\zeta \cdot \alpha\gamma\epsilon \cdot \delta\beta\epsilon = 0,$$

where

$$\alpha\beta\epsilon = \begin{vmatrix} P_\alpha & Q_\alpha & R_\alpha \\ P_\beta & Q_\beta & R_\beta \\ P_\epsilon & Q_\epsilon & R_\epsilon \end{vmatrix},$$

and substituting for  $P_\alpha$ , &c., their values, we have  $\alpha\beta\epsilon = w \cdot \mathfrak{p}_{\alpha\beta\epsilon}^2$ ; whence the foregoing result.

50. *Singularities.* The equation shows that

- (2) The line  $\alpha$  is a 2-tuple line:  $\mathfrak{p}_{\alpha\beta\epsilon}^2 = 0$ ,  $\mathfrak{p}_{\alpha\gamma\zeta}^2 = 0$ ,  $\mathfrak{p}_{\alpha\beta\zeta}^2 = 0$ ,  $\mathfrak{p}_{\alpha\gamma\epsilon}^2 = 0$ .
- (4) The line  $(\alpha, \beta, \epsilon, \zeta)$  is a simple line:  $\mathfrak{p}_{\alpha\beta\epsilon}^2 = 0$ ,  $\mathfrak{p}_{\alpha\beta\zeta}^2 = 0$ .
- (9) The quadriquadric  $\alpha\beta\epsilon.\gamma\delta\zeta = 0$  is a simple curve on the surface:  
for each point of the curve  $\mathfrak{p}_{\alpha\beta\epsilon}^2 = 0$ ,  $\mathfrak{p}_{\gamma\delta\zeta}^2 = 0$ .

It may be remarked that the surface meets the hyperboloid  $\mathfrak{p}_{\alpha\beta\epsilon}^2$  in

|                                               |             |                    |
|-----------------------------------------------|-------------|--------------------|
| lines $(\alpha, \beta, \epsilon)$             | each twice, | 6,                 |
| line $(\alpha, \beta, \epsilon, \gamma)$      | once,       | 2,                 |
| line $(\alpha, \beta, \epsilon, \delta)$      | once,       | 2,                 |
| line $(\alpha, \beta, \epsilon, \zeta)$       | once,       | 2,                 |
| curve $\alpha\beta\epsilon.\gamma\delta\zeta$ | twice,      | 4,                 |
|                                               |             | $2 \times 8 = 16.$ |

51. It might be thought that there should be on the surface some curve  $\alpha\beta\gamma\delta\epsilon\zeta$ , such as the cubic  $abcdef$  on the surface  $abcdef$ ; but I cannot find that this is so. The equation of the surface is satisfied if we have simultaneously ( $\lambda$  being arbitrary)

$$\mathfrak{p}_{\alpha\beta\epsilon}^2 \mathfrak{p}_{\alpha\gamma\zeta}^2 - \lambda \mathfrak{p}_{\alpha\beta\zeta}^2 \mathfrak{p}_{\alpha\gamma\epsilon}^2 = 0,$$

$$\lambda \mathfrak{p}_{\gamma\delta\epsilon}^2 \mathfrak{p}_{\delta\beta\zeta}^2 - \mathfrak{p}_{\gamma\delta\zeta}^2 \mathfrak{p}_{\delta\beta\epsilon}^2 = 0;$$

which equations represent quartic surfaces, the first of them having  $\alpha$  for a double line, and passing through the lines  $\beta, \gamma, \epsilon, \zeta$  ( $13 + 4 \times 5 = 33$  conditions, so that the equation of such a surface contains

only an arbitrary parameter  $\lambda$ ); and the second having  $\delta$  for a double line, and passing through the lines  $\beta, \gamma, \epsilon, \zeta$ . But I see no condition by which  $\lambda$  can be determined so as to have the same value in the two equations respectively. Of course, leaving it arbitrary, the two quartic surfaces intersect in the lines  $\beta, \gamma, \epsilon, \zeta$  and in a curve of the order 12 depending on the arbitrary value of  $\lambda$ , which curve lies on the surface  $\alpha\beta\gamma\delta\epsilon\zeta$ .

{Surface  $\alpha\beta\gamma\delta\epsilon\zeta$ .}

p. 132 [source PDF p. 503]

### The Excuboquartic $\alpha\beta\gamma, \delta\epsilon, a$ .

52. The notion is, that we have a fixed point  $a$ , two fixed lines  $\delta, \epsilon$ , and a singly infinite series of lines, or say the generating lines of a skew surface: each generating line determines, with the point  $a$ , a plane; and if in this plane we draw, meeting the lines  $\delta, \epsilon$ , a line to meet the generating line in a point  $P$ , then the locus of this point  $P$  is the curve about to be considered.

53. In the case in question, the singly infinite series of lines is that of the lines which meet each of the lines  $\alpha, \beta, \gamma$ , or say these are the generatrices of the hyperboloid  $\alpha\beta\gamma$ : the locus, or curve  $\alpha\beta\gamma, \delta\epsilon, a$ , is (as mentioned above) an excuboquartic. It is not necessary for the purpose of the memoir, but it is interesting to consider in conjunction therewith the excuboquartic arising in like manner from the directrices of the hyperboloid; it will appear that the two curves are the complete intersection of the quadric  $\alpha\beta\gamma$  by a quartic surface. Observe that the two curves are given as follows: viz. considering for the quadric  $\alpha\beta\gamma$  any tangent-plane through the point  $a$ , and drawing in this plane, to meet the lines  $\delta$  and  $\epsilon$ , a line, this meets the section of the quadric surface by the tangent-plane in two points, the locus of which is the aggregate of the two curves: viz. the section being a line-pair, the two points belong, one of them to a generatrix and the other to a directrix of the quadric surface.

54. It is convenient to take  $x = 0, y = 0$  for the equations of the line  $\delta$ ;  $z = 0, w = 0$  for those of the line  $\epsilon$ : for then, for any plane  $Ax + By + Cz + Dw = 0$ , the line in this plane and meeting the lines  $\delta$  and  $\epsilon$  has for its equations  $Ax + By = 0, Cz + Dw = 0$ ; or, what is the same thing, for the plane  $P = 0$  the equations of the line are  $P_{xy} = 0, P_{zw} = 0$ , where  $P_{xy}, P_{zw}$  denote the terms in  $x, y$  and in  $z, w$  respectively.

I take also  $x_0, y_0, z_0, w_0$  for the coordinates of the point  $a$ , and  $PS - QR = 0$  for the equation of the quadric surface,  $P, Q, R, S$  being given linear functions of  $(x, y, z, w)$ : we have then, say,  $P - \theta R = 0, Q - \theta S = 0$  for the equations of any generatrix, and  $P - \phi Q = 0, R - \phi S = 0$  for the equations of any directrix of the hyperboloid.

The equation of the plane through the point  $a$  and the generatrix  $P - \theta R = 0, Q - \theta S = 0$ , is clearly

$$(Q_0 - \theta S_0)(P - \theta R) - (P_0 - \theta R_0)(Q - \theta S) = 0;$$

so that for the line in this plane, meeting the lines  $\delta$  and  $\epsilon$ , we have

$$(Q_0 - \theta S_0)(P_{xy} - \theta R_{xy}) - (P_0 - \theta R_0)(Q_{xy} - \theta S_{xy}) = 0,$$

$$(Q_0 - \theta S_0)(P_{zw} - \theta R_{zw}) - (P_0 - \theta R_0)(Q_{zw} - \theta S_{zw}) = 0;$$

and joining thereto the equations

$$\theta = \frac{P}{R} = \frac{Q}{S} = \frac{P_{xy} + P_{zw}}{R_{xy} + R_{zw}} = \frac{Q_{xy} + Q_{zw}}{R_{xy} + R_{zw}},$$

(equivalent in all to three equations), the elimination of  $\theta$  gives the required curve: the equations thus are

$$PS - QR = 0,$$

$$(Q_0S - QS_0)(P_{xy}R - PR_{xy}) - (P_0R - PR_0)(Q_{xy}S - QS_{xy}) = 0,$$

p. 133 [source PDF p. 503]

or, as the second equation may also be written,

$$(Q_0S - QS_0)(P_{xy}R_{zw} - P_{zw}R_{xy}) - (P_0R - PR_0)(Q_{xy}S_{zw} - Q_{zw}S_{xy}) = 0;$$

viz. the second equation represents a cubic surface having upon it the lines  $(P = 0, R = 0)$  and  $(Q = 0, S = 0)$ : it therefore intersects the quadric  $PS - QR = 0$  in these two lines, and besides in an excuboquartic curve, which is the required locus.

55. Representing the determinants

$$\begin{vmatrix} P & Q & R & S \\ P_0 & Q_0 & R_0 & S_0 \end{vmatrix} \quad \text{by} \quad (a', b', c', f', g', h'),$$

viz.  $a' = QR_0 - Q_0R, \dots, f' = PS_0 - P_0S, \dots$ ; and

$$\begin{vmatrix} P_{xy} & Q_{xy} & R_{xy} & S_{xy} \\ P_{zw} & Q_{zw} & R_{zw} & S_{zw} \end{vmatrix} \quad \text{by} \quad (a, b, c, f, g, h),$$

viz.  $a = Q_{xy}R_{zw} - Q_{zw}R_{xy}, \dots$ ; so that  $(a', \dots)$  are linear functions,  $(a, \dots)$  quadric functions, of the coordinates; the equation of the cubic surface is

$$gb' - bg' = 0,$$

viz. the excuboquartic arising from the generatrices is the partial intersection of the quadric  $PS - QR = 0$  and the cubic  $gb' - g'b = 0$ ; the two surfaces besides intersecting in the lines  $(P = 0, R = 0)$  and  $(Q = 0, S = 0)$ .

It appears, in the same manner, that the excuboquartic arising from the directrices is the partial intersection of the quadric  $PS - QR = 0$  and the cubic  $hc' - ch' = 0$ ; the two surfaces besides intersecting in the lines  $(P = 0, Q = 0)$  and  $(R = 0, S = 0)$ .

56. But the elimination may be performed in a different manner, as follows: from the first two equations in  $\theta$ , multiplying by  $P_{zw}, -P_{xy}$  and adding, and so with  $Q_{zw}, -Q_{xy}$ , &c., we obtain

$$\begin{aligned} (Q_0 - \theta S_0)(-\theta b) - (P_0 - \theta R_0)(-c + \theta f) &= 0, \\ (Q_0 - \theta S_0)(c + \theta a) - (P_0 - \theta R_0)(\theta g) &= 0, \\ (Q_0 - \theta S_0)(-b) - (P_0 - \theta R_0)(a + \theta h) &= 0, \\ (Q_0 - \theta S_0)(f - \theta h) - (P_0 - \theta R_0)(g) &= 0. \end{aligned}$$

We then have

$$\theta = \frac{-c + \theta f}{a + \theta h} = \frac{c + \theta a}{f - \theta h},$$

or, what is the same thing,

$$h\theta^2 + (a - f)\theta + c = 0.$$

Using this equation, written in the form  $(a + \theta h)\theta = -c + \theta f$ , to transform the first or third of the four equations in  $\theta$ , we obtain

$$-aP_0 - bQ_0 - cR_0 + \theta(-hP_0 + fR_0 + bS_0) = 0;$$

and using the same equation, written in the form  $(f - \theta h)\theta = c + \theta a$ , to transform the second or fourth equation, we obtain

$$gP_0 - fQ_0 + cS_0 + \theta(hQ_0 - gR_0 + aS_0) = 0;$$

p. 134 [source PDF p. 503]



and hence, eliminating  $\theta$ , we obtain

$$(hQ_0 - gR_0 + aS_0)(-aP_0 - bQ_0 - cR_0) - (-hP_0 + fR_0 + bS_0)(gP_0 - fQ_0 + cS_0) = 0,$$

which, as being of the second order in  $(a, \dots)$ , represents a quartic surface. The equation remains unaltered by the interchange of  $Q, R$ , and the consequent interchanges among  $(a, b, c, f, g, h)$ : hence the quartic surface contains not only the excuboquartic arising from the generatrices, but also that arising from the directrices; and these two curves are the complete intersection of the quartic by the quadric  $PS - QR = 0$ .

57. I obtain this same result also as follows. Consider a point  $(P_1, Q_1, R_1, S_1)$  on the quadric surface;  $P_1S_1 - Q_1R_1 = 0$ ; the tangent plane at the point is

$$PS_1 - QR_1 - RQ_1 + SP_1 = 0;$$

and if this passes through the point  $a$ , then

$$P_0S_1 - Q_0R_1 - R_0Q_1 + S_0P_1 = 0.$$

The line which in the tangent-plane meets the lines  $\delta, \epsilon$  is given, as before, by the equations

$$P_{xy}S_1 - Q_{xy}R_1 - R_{xy}Q_1 + S_{xy}P_1 = 0,$$

$$P_{zw}S_1 - Q_{zw}R_1 - R_{zw}Q_1 + S_{zw}P_1 = 0.$$

Remembering the significations of  $(a, \dots)$ , the last three equations give

$$\begin{aligned} S_1 : R_1 : -Q_1 : -P_1 &= hQ_0 - gR_0 + aS_0 \\ &: -hP_0 + fR_0 + bS_0 \\ &: gP_0 - fQ_0 + cS_0 \\ &: -aP_0 - bQ_0 - cR_0; \end{aligned}$$

and substituting these values in  $S_1P_1 - Q_1R_1 = 0$ , we have the above equation of the quadric surface.

58. Or again, changing the notation, I take the equation of the quadric surface to be

$$(a, b, c, d, f, g, h, l, m, n)(x, y, z, w)^2 = 0.$$

A tangent-plane hereof is

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

where  $\xi, \eta, \zeta, \omega$  are any quantities satisfying the relation

$$(A, B, C, D, F, G, H, L, M, N)(\xi, \eta, \zeta, \omega)^2 = 0,$$

the capitals denoting the inverse coefficients.

Supposing that the tangent-plane passes through a fixed point  $a$ , coordinates  $(\alpha, \beta, \gamma, \delta)$ , we have

$$\alpha\xi + \beta\eta + \gamma\zeta + \delta\omega = 0;$$

*p. 135 [source PDF p. 503]*

and if the equations of the lines  $\delta, \epsilon$  are as before ( $x = 0, y = 0$ ) and ( $z = 0, w = 0$ ); then for the line in the tangent-plane meeting the lines  $\delta, \epsilon$ , we have

$$\xi x + \eta y = 0, \quad \zeta z + \omega w = 0.$$

These last equations may be represented by

$$\xi = ly, \quad \eta = -lx, \quad \zeta = mw, \quad \omega = -mz;$$

and, substituting these values, we have

$$(A, \dots)(ly, -lx, mw, -mz)^2 = 0, \quad (\alpha, \dots)(ly, -lx, mw, -mz)^1 = 0,$$

that is

$$(Ay^2 - 2Hxy + Bx^2, -Fwx + Gyw - Lyz + Mxz, Cw^2 - 2Nwz + Dz^2)(l, m)^2 = 0,$$

and

$$(\alpha y - \beta x, \gamma w - \delta z)(l, m) = 0.$$

Whence, eliminating  $l, m$ , we have the quartic equation

$$(Ay^2 - 2Hxy + Bx^2, -Fwx + Gyw - Lyz + Mxz, Cw^2 - 2Nwz + Dz^2)(\gamma w - \delta z, \beta x - \alpha y)^2 = 0.$$

### Further Investigation as to the Surface $ab\alpha\beta\gamma\delta$ .

59. The theorem that in the surface  $ab\alpha\beta\gamma\delta$ , the equation of which is

$$\text{Norm}\{\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}}\mathfrak{p}_{\beta\gamma\delta}^2 - \sqrt{\mathfrak{p}_{a\beta}\mathfrak{p}_{b\beta}}\mathfrak{p}_{\gamma\delta\alpha}^2 + \sqrt{\mathfrak{p}_{a\gamma}\mathfrak{p}_{b\gamma}}\mathfrak{p}_{\delta\alpha\beta}^2 - \sqrt{\mathfrak{p}_{a\delta}\mathfrak{p}_{b\delta}}\mathfrak{p}_{\alpha\beta\gamma}^2\} = 0,$$

the lines  $(ab, \alpha, \beta, \gamma)$  are tacnodal, each sheet touching along the line the quadric  $\mathfrak{p}_{\alpha\beta\gamma}^2$ , may be proved in a different manner by investigating the intersection of the surface with the quadric  $\mathfrak{p}_{\alpha\beta\gamma}^2$ .

For this purpose take the equation of the quadric to be  $yz - xw = 0$ ; the equations of the lines  $\alpha, \beta, \gamma$  will be

$$\begin{pmatrix} z - \lambda_\alpha w = 0 \\ x - \lambda_\alpha y = 0 \end{pmatrix}, \quad \begin{pmatrix} z - \lambda_\beta w = 0 \\ x - \lambda_\beta y = 0 \end{pmatrix}, \quad \begin{pmatrix} z - \lambda_\gamma w = 0 \\ x - \lambda_\gamma y = 0 \end{pmatrix};$$

and we may write  $(a, b, c, f, g, h)$  for the coordinates of the line  $\delta$ . The equation of the surface will be

$$\begin{aligned} \text{Norm} \left\{ \Sigma \left[ \pm \sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}}(\lambda_\beta - \lambda_\gamma) \begin{pmatrix} (a-f)xz - (\lambda_\beta + \lambda_\gamma)yz + \lambda_\beta\lambda_\gamma yw \\ (b-g)\lambda_\beta\lambda_\gamma(yz - xw) \\ c(z - \lambda_\beta w)(z - \lambda_\gamma w) \\ h(x - \lambda_\beta y)(x - \lambda_\gamma z) \end{pmatrix} \right] \right. \\ \left. - \sqrt{\mathfrak{p}_{a\delta}\mathfrak{p}_{b\delta}}(\lambda_\beta - \lambda_\gamma)(\lambda_\gamma - \lambda_\alpha)(\lambda_\alpha - \lambda_\beta)(yz - xw) \right\}. \end{aligned}$$

*p. 136 [source PDF p. 503]*

Here  $\Sigma$  denotes the sum of the three terms obtained by the cyclical interchange of  $\alpha, \beta, \gamma$ ; and

$$\mathfrak{p}_{a\alpha} = (z_a - \lambda w_a)(x - \lambda y) - (x_a - \lambda y_a)(z - \lambda w),$$

$$\mathfrak{p}_{b\alpha} = (z_b - \lambda w_b)(x - \lambda y) - (x_b - \lambda y_b)(z - \lambda w);$$

$\lambda$  here standing for  $\lambda_\alpha$ ; and similarly for  $\mathfrak{p}_{a\beta}$ , &c.

60. To obtain the intersection with  $xw - yz = 0$ , writing  $w = \frac{yz}{x}$ , then

$$\mathfrak{p}_{a\alpha} = \left[ z_a - \lambda w_a - \frac{z}{x}(x_a - \lambda y_a) \right] (x - \lambda y), \quad (\lambda = \lambda_\alpha),$$

$$\mathfrak{p}_{b\alpha} = \left[ z_b - \lambda w_b - \frac{z}{x}(x_b - \lambda y_b) \right] (x - \lambda y);$$

or say

$$\sqrt{\mathfrak{p}_{a\alpha}\mathfrak{p}_{b\alpha}} = \sqrt{M_\alpha}(x - \lambda_\alpha y).$$

Also the expression in braces becomes

$$= \{(a-f)\frac{z}{x} + c\frac{z^2}{x^2} + h\}(x - \lambda_\beta y)(x - \lambda_\gamma y);$$

so that the norm in question is

$$\text{Norm } \Sigma \sqrt{M_\alpha}(\lambda_\beta - \lambda_\gamma) \{(a-f)\frac{z}{x} + c\frac{z^2}{x^2} + h\}(x - \lambda_\alpha y)(x - \lambda_\beta y)(x - \lambda_\gamma y);$$

or say

$$\text{Norm } \Sigma \sqrt{M_\alpha}(\lambda_\beta - \lambda_\gamma) \{hx^2 + (a-f)zx + cz^2\}(x - \lambda_\alpha y)(x - \lambda_\beta y)(x - \lambda_\gamma y);$$

where  $M_\alpha$  is now considered to stand for

$$\{(z_ax - zx_a) - \lambda(w_ax - y_az)\}\{(z_bx - zx_b) - \lambda(w_bx - y_bz)\}.$$

Observing that the norm was originally the product of 8 factors, this breaks up into

$$\{hx^2 + (a-f)zx + cz^2\}^8 \{(x - \lambda_\alpha y)(x - \lambda_\beta y)(x - \lambda_\gamma y)\}^8 = 0,$$

and

$$\text{Norm}^2 \sqrt{M_\alpha}(\lambda_\beta - \lambda_\gamma) = 0,$$

where the new norm is the product of 4 factors.

61. Writing for greater convenience  $\lambda, \mu, \nu$  in place of  $\lambda_\alpha, \lambda_\beta, \lambda_\gamma$ , and observing that  $M_\alpha$  is a quadric function of  $\lambda_\alpha$ , that is of  $\lambda$ , the last-mentioned norm is

$$\text{Norm } \sqrt{A + B\lambda + C\lambda^2}(\mu - \nu),$$

which is easily seen to be

$$= (4AC - B^2)(\mu - \nu)^2(\nu - \lambda)^2(\lambda - \mu)^2;$$

or writing for a moment

$$(A + B\lambda + C\lambda^2) = (P - Q\lambda)(P' - Q'\lambda),$$

*p. 137 [source PDF p. 503]*

whence

$$A = PP', \quad B = -(PQ' + P'Q), \quad C = QQ';$$

then

$$4AC - B^2 = -(PQ' - P'Q)^2;$$

and we have

$$P, Q = z_ax - zx_a, \quad w_ax - y_az,$$

$$P', Q' = z_bx - zx_b, \quad w_bx - y_bz,$$

whence

$$\begin{aligned} PQ' - P'Q &= (z_aw_b - z_bw_a)x^2 \\ &\quad + [y_az_b - y_bz_a - (x_aw_b - x_bw_a)]xz \\ &\quad + (x_ay_b - x_by_a)z^2; \end{aligned}$$

viz. if  $(a, b, c, f, g, h)$  are the coordinates of the line  $ab$ , this is

$$= hx^2 + (a-f)xz + cz^2.$$

Hence, omitting the constant factor  $(\mu - \nu)^4(\nu - \lambda)^4(\lambda - \mu)^4$

$$\{\text{that is } (\lambda_\beta - \lambda_\gamma)^4(\lambda_\gamma - \lambda_\alpha)^4(\lambda_\alpha - \lambda_\beta)^4\},$$

the foregoing equation  $\text{Norm}^2 = 0$  becomes

$$[hx^2 + (a - f)xz + cz^2]^4 = 0,$$

and the intersections of the quadric with the surface are obtained by combining the equation  $xw - yz = 0$  with the several equations

$$\{hx^2 + (a - f)xz + cz^2\}^8 = 0, \quad \{(x - \lambda_\alpha y)(x - \lambda_\beta y)(x - \lambda_\gamma y)\}^8 = 0,$$

$$\{hx^2 + (a - f)zx + cz^2\}^4 = 0;$$

viz. these are

|                                         |              |     |
|-----------------------------------------|--------------|-----|
| lines $(\alpha, \beta, \gamma, \delta)$ | each 8 times | 16, |
| line $(x = 0, z = 0)$                   | 16 times     | 16, |
| lines $\alpha, \beta, \gamma$           | each 8 times | 24, |
| line $(x = 0, y = 0)$                   | 24 times     | 24, |
| lines $[ab, \alpha, \beta, \gamma]$     | each 4 times | 8,  |
| line $(x = 0, z = 0)$                   | 8 times      | 8,  |
| $(16 + 24 + 8) \times 2 = 48 + 48.$     |              |     |

But it is clear that the lines  $(x = 0, y = 0)$  and  $(x = 0, z = 0)$  are introduced by the process of elimination, and are no part of the intersection. The complete intersection consists of the lines  $(\alpha, \beta, \gamma, \delta)$  each 8 times, the lines  $(\alpha, \beta, \gamma)$  each 8 times, and the lines  $[ab, \alpha, \beta, \gamma]$  each 4 times. But the last-mentioned lines being only double lines on the surface, this means that the two sheets each touch the quadric surface, or that the lines are tacnodal.

*p. 138 [source PDF p. 504]*

[504]

### On the Mechanical Description of Certain Sextic Curves.

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THE curves in question might be taken to be those described by a point  $C$  rigidly connected with points  $A$  and  $B$ , each of which describes a circle: but the construction is considered under a somewhat more general form. I consider a quadrilateral, the sides of which are  $a, b, c, d$ , and the inclinations of these to a fixed line  $\alpha, \beta, \gamma, \delta$ . This being so, if  $a, b, c, d$ , and one of the angles, say  $\delta$ , are constant, then we have between the three variable angles the relations

$$a \cos \alpha + b \cos \beta + c \cos \gamma + d \cos \delta = 0,$$

$$a \sin \alpha + b \sin \beta + c \sin \gamma + d \sin \delta = 0,$$

giving rise to a single relation between any two of the variable angles; and we consider a curve such that the coordinates  $x, y$  of any point thereof are given linear functions of the sines and cosines of the three variable angles, or, what is the same thing, of the sines and cosines of any two of these angles. We thus unite together what would otherwise be distinct cases; for everything is symmetrical in regard to the sides  $a, b, c$  and the corresponding variable angles  $\alpha, \beta, \gamma$ , irrespectively of the order of succession of these sides: and we can thus, in the discussion of the curve, employ any two at pleasure, say  $\alpha, \beta$ , of the variable angles, without determining whether the sides  $a, b$  are contiguous or opposite.

Eliminating, then, the variable angle  $\gamma$ , we obtain between  $\alpha, \beta$  a relation which, if we write therein  $\tan \frac{1}{2}\alpha = u$ ,  $\tan \frac{1}{2}\beta = v$ , takes the form

$$(*\chi u, 1)^2(v, 1)^2 = 0;$$

viz. either of the variables  $u, v$  is expressible rationally in terms of the other of them and of the root of a quartic function thereof; say  $v$  is a rational function of  $u$  and  $\sqrt{U}$ .

*p. 139 [source PDF p. 504]*

And hence a curve for which the coordinates  $x, y$  are rational functions of  $u, v$ , is a curve having a deficiency  $D = 1$ , or, what is the same thing, having a number of dps. less by unity than the maximum number  $\{= \frac{1}{2}(n-1)(n-2)$ , if  $n$  be the order of the curve $\}$ .

It will further appear that the relation  $(*\chi u, 1)^2(v, 1)^2 = 0$  is satisfied by the values  $u = v = i$  and  $u = v = -i$  (if, as usual,  $i = \sqrt{-1}$ ).

In the curve in question, the coordinates  $(x, y)$  are given linear functions of the sines and cosines of  $\alpha, \beta$ ; and if we make the curve meet an arbitrary line  $Ax + By + C = 0$ , we obtain between the sines and cosines of  $\alpha, \beta$  a linear relation which, substituting therein the expressions in terms of  $u, v$ , takes the form

$$(*\chi u, 1)^2(1 + v^2) + (*\chi v, 1)^2(1 + u^2) = 0,$$

viz. this is a relation of the form

$$(\dagger\chi u, 1)^2(v, 1)^2 = 0,$$

such that it is satisfied by the four sets of values  $u = \pm i, v = \pm i$ , and therefore in particular by the values  $u = v = i$  and  $u = v = -i$ .

Hence, considering the intersections of the curve by the arbitrary line, the values of  $(u, v)$  are given by the two equations

$$(*\chi u, 1)^2(v, 1)^2 = 0, \quad (\dagger\chi u, 1)^2(v, 1)^2 = 0;$$

these, regarding for a moment  $u, v$  as ordinary rectangular coordinates, represent each of them a quartic curve having two dps. at infinity on the axes  $u = 0, v = 0$  respectively: each of these points reckons therefore as 4 intersections, and the number of the remaining intersections therefore is  $4 \cdot 4 - 2 \cdot 4 = 8$ . But, by what precedes, the two quartic curves have also in common the points  $u = v = i$  and  $u = v = -i$ ; and rejecting these, there remain  $8 - 2 = 6$  intersections.

The conclusion is, that the curve is a sextic curve of deficiency 1, that is, having 9 dps. The reasoning may be presented under a slightly different form as follows: regarding  $u, v$  as coordinates, we have the curve  $(*\chi u, 1)^2(v, 1)^2 = 0$ , a binodal quartic curve, and having therefore the deficiency 1; the curve passes, as above-mentioned, through the points  $u = v = i$  and  $u = v = -i$ . The required curve is obtained as a transformation of the quartic curve by formulae of the form  $x : y : z (= 1) = P : Q : R$ , where  $P, Q$ , and  $R \{= (1 + u^2)(1 + v^2)\}$  are quartic functions of the coordinates  $u, v$ , such that  $P = 0, Q = 0, R = 0$  are each of them a quartic curve passing twice through each of the nodes and once through each of the before-mentioned points,  $(u = v = i)$  and  $(u = v = -i)$ , of the binodal quartic curve. Hence the curve in question is a curve of the order  $4 \cdot 4 - 2 \cdot 4 - 2 \cdot 1 = 6$ , and having the same deficiency as the binodal quartic, that is, the deficiency is = 1.

I observe that the sextic curve does not, in general, pass through the circular points at infinity, but it intersects the line at infinity in three distinct pairs of points; one of these, or all three of them, (but not two pairs only,) may coincide with the circular points at infinity, the circular points at infinity being, in the latter case, triple points, or the curve being tricircular: this will appear presently.

To obtain the foregoing equation  $(*\chi u, 1)^2(v, 1)^2 = 0$ , the elimination of  $\gamma$  gives

$$(a \cos \alpha + b \cos \beta + d \cos \delta)^2 + (a \sin \alpha + b \sin \beta + d \sin \delta)^2 = c^2,$$

*p. 140 [source PDF p. 504]*

that is

$$a^2 + b^2 - c^2 + d^2 + 2ab \cos(\alpha - \beta) + 2ad \cos(\alpha - \delta) + 2bd \cos(\beta - \delta) = 0;$$

or, substituting herein the values

$$\cos \alpha = \frac{1 - u^2}{1 + u^2}, \quad \sin \alpha = \frac{2u}{1 + u^2}, \quad \cos \beta = \frac{1 - v^2}{1 + v^2}, \quad \sin \beta = \frac{2v}{1 + v^2},$$

this is easily found to be  $0 =$

|       | 1                                           | $2u$              | $u^2$                                       |
|-------|---------------------------------------------|-------------------|---------------------------------------------|
| 1     | $(a+b)^2 - 2(a+b)d \cos \delta + d^2 - c^2$ | $2ad \sin \delta$ | $(a-b)^2 + 2(a-b)d \cos \delta + d^2 - c^2$ |
| $2v$  | $2bd \sin \delta$                           | $2ab$             | $2bd \sin \delta$                           |
| $v^2$ | $(a-b)^2 - 2(a-b)d \cos \delta + d^2 - c^2$ | $2ad \sin \delta$ | $(a+b)^2 + 2(a+b)d \cos \delta + d^2 - c^2$ |

Or writing, for greater convenience,  $\delta = 0$ , that is, taking  $\alpha, \beta, \gamma$  to be the inclinations of the sides  $a, b, c$  to the fixed side  $d$ , this is  $0 =$

|       | 1                 | $2u$  | $u^2$             |
|-------|-------------------|-------|-------------------|
| 1     | $(a+b-d)^2 - c^2$ | 0     | $(a-b+d)^2 - c^2$ |
| $2v$  | 0                 | $2ab$ | 0                 |
| $v^2$ | $(a-b-d)^2 - c^2$ | 0     | $(a+b+d)^2 - c^2$ |

or say this is

$$(A + Bu^2) + 8abuv + v^2(C + Du^2) = 0,$$

where, writing for shortness

$$\begin{aligned} a + d - b - c &= \lambda, & -a + b + c + d &= \lambda', \\ b + d - c - a &= \mu, & -b + c + d + a &= \mu', \\ c + d - a - b &= \nu, & -c + d + a + b &= \nu', \\ a + b + c + d &= \sigma, & -d + a + b + c &= \rho', \end{aligned}$$

then

$$A = -\nu\rho', \quad B = \lambda\mu', \quad C = \lambda'\mu, \quad D = \nu'\sigma.$$

We at once verify that the equation is satisfied by  $u = v = \pm i$ , viz. this will be so if only

$$A - B - C + D - 8ab = 0,$$

*p. 141 [source PDF p. 504]*

and we have

$$A + D = 2(a+b)^2 + 2d^2 - 2c^2, \quad B + C = 2(a-b)^2 + 2d^2 - 2c^2;$$

whence the relation in question.

Writing  $u = i$ , we have

$$\{v(C - D) + 4abi\}^2 = -16a^2b^2 - (A - B)(C - D) = -\frac{1}{4}(A - B + C - D)^2,$$

whence

$$v(C - D) = \pm \frac{1}{2}i\{\mp 8ab + A - B + C - D\} = i(C - D) \quad \text{or} \quad -i(A - B);$$

so that, corresponding to  $u = i$ , we have the values

$$v = i, \quad v = -i \frac{A - B}{C - D};$$

and similarly, corresponding to  $u = -i$ , the values

$$v = -i, \quad v = i \frac{A - B}{C - D}.$$

And in like manner, corresponding to  $v = i$ , we have the values

$$u = i, \quad u = -i \frac{A - C}{B - D};$$

and to  $v = -i$ , the values

$$u = -i, \quad u = i \frac{A - C}{B - D}.$$

It is easy to show that, if  $u = i + \varepsilon$ ,  $v = i + \zeta$  are the values consecutive to  $u = v = i$ , then  $a\varepsilon + b\zeta = 0$ : in fact, substituting the foregoing values in the relation between  $u, v$ , and writing for  $8ab$  its value,  $= A - B - C + D$ , we have

$$A + B(-1 + 2i\varepsilon) + (A - B - C + D)\{-1 + i(\varepsilon + \zeta)\} + C(-1 + 2i\zeta) + D\{1 - 2i(\varepsilon + \zeta)\} = 0,$$

which is

$$= \varepsilon(A + B - C - D) + \zeta(A - B + C - D) = 0;$$

or finally  $a\varepsilon + b\zeta = 0$ . And similarly, if  $u = -i + \varepsilon$ ,  $v = -i + \zeta$  are the values consecutive to  $u = v = -i$ , then we have the same relation  $a\varepsilon + b\zeta = 0$ .

The points at infinity on the sextic curve are those for which  $1 + u^2$  or  $1 + v^2$ , or each of these, is  $= 0$ ; viz. the values of  $u, v$  for the six points are

$$\begin{array}{llllll} u = i + \varepsilon, & -i - \varepsilon, & i, & -i, & -i \frac{A - C}{B - D}, & i \frac{A - C}{B - D}, \\ v = i + \zeta, & -i - \zeta, & -i \frac{A - B}{C - D}, & i \frac{A - B}{C - D}, & i, & -i, \end{array}$$

where, instead of  $u = v = i$  and  $u = v = -i$ , I have written down the consecutive values of  $u, v$ , and as before  $\varepsilon, \zeta$  are infinitesimals such that  $a\varepsilon + b\zeta = 0$ .

Suppose that the coordinates  $x, y$  of a point on the sextic curve are

$$x = a \cos \alpha + c \sin \alpha + b \cos \beta + d \sin \beta,$$

$$y = a' \cos \alpha + c' \sin \alpha + b' \cos \beta + d' \sin \beta;$$

*p. 142 [source PDF p. 206]*

then, if  $u = \pm(i + \varepsilon)$ ,

$$\cos \alpha = \frac{-i}{\varepsilon}, \quad \sin \alpha = \frac{\pm 1}{\varepsilon},$$

and similarly if  $v = \pm(i + \zeta)$ , then

$$\cos \beta = \frac{-i}{\zeta}, \quad \sin \beta = \frac{\pm 1}{\zeta}.$$

Hence the points at infinity of the sextic curve are as follows:

$$\begin{array}{ll} 1^\circ. & u = \pm(i + \varepsilon), \quad v \neq \pm(i + \zeta), \\ & x = \frac{-ai \pm c}{\varepsilon}, \quad y = \frac{-a'i \pm c'}{\varepsilon}, \quad \text{first pair of points;} \\ 2^\circ. & v = \pm(i + \zeta), \quad u \neq \pm(i + \varepsilon), \\ & x = \frac{-bi \pm d}{\zeta}, \quad y = \frac{-b'i \pm d'}{\zeta}, \quad \text{second pair of points;} \\ 3^\circ. & u = \pm(i + \varepsilon), \quad v = \pm(i + \zeta), \\ & x = \frac{-ai \pm c}{\varepsilon} + \frac{-bi \pm d}{\zeta}, \quad y = \frac{-a'i \pm c'}{\varepsilon} + \frac{-b'i \pm d'}{\zeta}; \\ & a\varepsilon + b\zeta = 0, \quad \text{as above, third pair of points.} \end{array}$$

which six points are in general distinct from each other, and from the circular points at infinity.

The foregoing values of  $x, y$  may be said to be “circular quoad  $\alpha$ ,” if  $a = c'$ ,  $a' = -c$ ; and similarly to be “circular quoad  $\beta$ ,” if  $b = d'$ ,  $b' = -d$ .

And we see at once that if the values are circular quoad  $\alpha$ , then the first pair of points coincide with the circular points at infinity; and that, in like manner, if the values are circular quoad  $\beta$ , then the second pair of points coincide with the circular points at infinity; but if the values are circular quoad  $\alpha$  and  $\beta$  respectively, then each of the three pairs of points coincides with the circular points at infinity: so that these are then triple points on the curve; or the curve is tricircular, having besides the two triple points, 3 dps.

The relation between  $u, v$  gives

$$\{v(C + Du^2) + 4abu\}^2 = 16a^2b^2 \cdot u^2 - (A + Bu^2)(C + Du^2),$$

and it thus appears that if any one of the functions  $A, B, C, D$  is  $= 0$ , the function under the radical sign is a mere linear function of  $u^2$ , say it is  $L + Mu^2$ ; introducing a new parameter  $\theta$  such that

$$u = \sqrt{\frac{L}{M}} \frac{2\theta}{1 + \theta^2}, \quad \sqrt{L + Mu^2} = \sqrt{L} \frac{1 + \theta^2}{1 - \theta^2},$$

and consequently  $u, v$  are each of them a rational function of  $\theta$ . Hence, when any one of the relations in question is satisfied, or say, when  $a + d = b + c$ ,  $b + d = c + a$ , or  $c + d = a + b$ , the curve becomes unicursal: there is no diminution of the order, and the curve is consequently a unicursal sextic, or sextic with 10 dps.

*p. 143 [source PDF p. 207]*

It would at first sight appear that the curve might become unicursal in a different manner; viz. it would be unicursal if

$$16a^2b^2u^2 - (A + Bu^2)(C + Du^2)$$

was a perfect square; but this is only the case when one of the four sides  $a, b, c, d$  is  $= 0$ . The condition in fact is

$$AD + BC - 16a^2b^2 = 2\sqrt{ABCD};$$

that is

$$\lambda\lambda'\mu\mu' - \nu\nu'\rho'\sigma - 16a^2b^2 = 2\sqrt{-\lambda\lambda'\mu\mu'\rho'\sigma};$$

where, putting for shortness  $M = d^2 - a^2 - b^2 - c^2$ , we have

$$\begin{aligned} \lambda\lambda' &= M - 2bc + 2ca + 2ab, \\ \mu\mu' &= M + 2bc - 2ca + 2ab, \\ \nu\nu' &= M + 2bc + 2ca - 2ab, \\ -\rho'\sigma &= M - 2bc - 2ca - 2ab; \end{aligned}$$

and thence

$$\begin{aligned} \lambda\lambda'\mu\mu' &= M^2 - 4b^2c^2 - 4c^2a^2 + 4a^2b^2 + 4abM + 8c^2ab, \\ -\nu\nu'\rho'\sigma &= M^2 - 4b^2c^2 - 4c^2a^2 + 4a^2b^2 - 4abM - 8c^2ab, \end{aligned}$$

and the equation thus becomes

$$M^2 - 4b^2c^2 - 4c^2a^2 - 4a^2b^2 = \sqrt{(M^2 - 4b^2c^2 - 4c^2a^2 + 4a^2b^2)^2 - 16a^2b^2(M + 2c^2)^2}.$$

viz. putting for a moment  $X = M^2 - 4c^2(a^2 + b^2)$ , this is

$$(X - 4a^2b^2)^2 = (X + 4a^2b^2)^2 - 16a^2b^2(M + 2c^2)^2,$$

that is

$$16a^2b^2\{X - (M + 2c^2)^2\} = 0;$$



or, substituting for  $X$  its value, the equation is

$$64a^2b^2c^2(M + a^2 + b^2 + c^2) = 0,$$

that is  $a^2b^2c^2d^2 = 0$ .

We may have simultaneously  $1^\circ$ ,  $a = d$ ,  $b = c$ ;  $2^\circ$ ,  $b = d$ ,  $a = c$ ;  $3^\circ$ ,  $c = d$ ,  $a = b$ ; the three cases are really equivalent, but the results present themselves in different forms.

$1^\circ$ . Here  $A = 0$ ,  $B = 4a(a - b)$ ,  $C = 0$ ,  $D = 4a(a + b)$ ; the relation between  $u, v$  contains the factor  $u$ , and throwing this out, and also the constant factor  $4a$ , it is

$$u[(a - b) + (a + b)v^2] + 2bv = 0,$$

viz.  $u$  is given as a rational function of  $v$ .

*p. 144 [source PDF p. 208]*

$2^\circ$ . Here  $A = 0$ ,  $B = 0$ ,  $C = 4b(b - a)$ ,  $D = 4b(b + a)$ ; the equation contains the factor  $v$ , and throwing out this and also the constant factor  $4b$ , the equation is

$$v[(b - a) + (b + a)u^2] + 2au = 0,$$

viz.  $v$  is given as a rational function of  $u$ .

$3^\circ$ . Here  $A = 4a(a - c)$ ,  $B = 0$ ,  $C = 0$ ,  $D = 4a(a + c)$ ; or, dividing by  $4a$ , the equation is

$$(a - c) + 2auv + (a + c)u^2v^2 = 0;$$

viz. this is

$$(uv + 1)[(a + c)uv + a - c] = 0,$$

which may be reduced to

$$(a + c)uv + a - c = 0,$$

giving  $u$  or  $v$  each a rational function of the other.

I do not discuss the theory in detail, but only remark that in each case there is a conic thrown off, and that in place of the sextic we have a unicursal (or trinodal) quartic curve.

*p. 145 [source PDF p. 209]*

505.

## ON THE SURFACES DIVISIBLE INTO SQUARES BY THEIR CURVES OF CURVATURE

*[From the Proceedings of the London Mathematical Society, vol. iv. (1871–1873), pp. 120, 121. Read June 13, 1872.]*

Professor Cayley gave an account of an investigation recently communicated by him to the Academy of Sciences at Paris. The fundamental theorem is that, if the coordinates  $x, y, z$  of a point on a surface are expressed as functions of two parameters  $p, q$  (such expressions, of course replacing the equation of the surface); and if these parameters are such that  $p = \text{const.}$ ,  $q = \text{const.}$  are the equations of the two sets of curves of curvature respectively; then (writing for shortness

$$\frac{dx}{dp} = x_1, \quad \frac{dx}{dq} = x_2, \quad \frac{d^2x}{dp^2} = x_3, \quad \frac{d^2x}{dp dq} = x_4, \quad \frac{d^2x}{dq^2} = x_5,$$

and the like for  $y, z$ ), the coordinates  $x, y, z$ , considered always as functions of  $p, q$ , satisfy the equations

$$x_1x_2 + y_1y_2 + z_1z_2 = 0, \quad \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_4 & y_4 & z_4 \end{vmatrix} = 0.$$

The last equation is equivalent to

$$\begin{aligned}x_4 + Ax_1 + Bx_2 &= 0, \\y_4 + Ay_1 + By_2 &= 0, \\z_4 + Az_1 + Bz_2 &= 0;\end{aligned}$$

*p. 146 [source PDF p. 210]*

and if in the notation of Gauss we write

$$x_1^2 + y_1^2 + z_1^2 = E, \quad x_2^2 + y_2^2 + z_2^2 = G,$$

then adding the equations multiplied by  $x_1, y_1, z_1$  respectively, and also adding the equations multiplied by  $x_2, y_2, z_2$  respectively, we find

$$A = -\frac{1}{2} \frac{1}{E} \frac{dE}{dq}, \quad B = -\frac{1}{2} \frac{1}{G} \frac{dG}{dp};$$

and the equations thus become

$$2x_4 - \frac{1}{E} \frac{dE}{dq} x_1 - \frac{1}{G} \frac{dG}{dp} x_2 = 0, \quad \&c. \quad \&c. \quad \&c.,$$

which, in fact, agree with the equations (10 bis) in Lamé's "Leçons sur les coordonnées curvilignes," Paris (1859), p. 89. The surface will be divisible into squares if only  $E : G$  is the quotient of a function of  $p$  by a function of  $q$ , or say if

$$E = \Theta P, \quad G = \Theta Q,$$

where  $\Theta$  is any function of  $(p, q)$ , but  $P$  and  $Q$  are functions of  $p$  and  $q$  respectively; we then have

$$\frac{1}{E} \frac{dE}{dq} = \frac{1}{\Theta} \frac{d\Theta}{dq}, \quad \frac{1}{G} \frac{dG}{dp} = \frac{1}{\Theta} \frac{d\Theta}{dp},$$

and the equations for  $x, y, z$  are

$$2x_4 - \frac{1}{\Theta} \frac{d\Theta}{dq} x_1 - \frac{1}{\Theta} \frac{d\Theta}{dp} x_2 = 0, \quad \&c. \quad \&c. \quad \&c.,$$

viz.  $x, y, z$  being functions of  $p, q$  such that  $x_1x_2 + y_1y_2 + z_1z_2 = 0$ , and which besides satisfy these equations, or say which each of them satisfy the equation

$$2u_4 - \frac{1}{\Theta} \frac{d\Theta}{dq} u_1 - \frac{1}{\Theta} \frac{d\Theta}{dp} u_2 = 0,$$

then the values of  $x, y, z$  in terms of  $(p, q)$  determine a surface which has the property in question.

*p. 147 [source PDF p. 211]*

506.

## ON THE MECHANICAL DESCRIPTION OF A CUBIC CURVE

*[From the Proceedings of the London Mathematical Society, vol. iv. (1871-1873), pp. 175-178. Read November 14, 1872.]*

If the coordinates  $x, y$  of a point on a curve are rational functions of  $\sin \phi, \cos \phi, \sqrt{1 - k^2 \sin^2 \phi}$ , the curve has the deficiency 1, and conversely in any curve of deficiency 1 the coordinates  $x, y$  can be thus expressed in terms of the parameter  $\phi$ . Hence writing  $\sin \theta = k \sin \phi$ , the coordinates will be rational functions of  $\sin \phi, \cos \phi, \cos \theta$ , or say of  $\sin \phi, \cos \phi, \sin \theta, \cos \theta$ ; and for the mechanical representation of

the relation  $k \sin \phi = \sin \theta$ , we require only a rod  $OA$  rotating about the fixed point  $O$ , and connected with it by a pin at  $A$ , a rod  $AB$ , the other extremity of which,  $B$ , moves in a fixed line  $Ox$ . The curve most readily obtained by such an arrangement is that described by a point  $C$  rigidly connected with the rod  $AB$ ; this is however a quartic curve (with two dps., since its deficiency is  $= 1$ ). I first considered the cubic curve

$$xy - 1 = \sqrt{(1 - x^2)(1 - k^2 x^2)},$$

or say

$$xy - 1 = -\sqrt{(1 - x^2)(1 - k^2 x^2)};$$

writing herein  $x = \sin \phi$ , and as before  $k \sin \phi = \sin \theta$ , we have then  $y \sin \phi = 1 - \cos \theta \cos \phi$ ; which values may be written

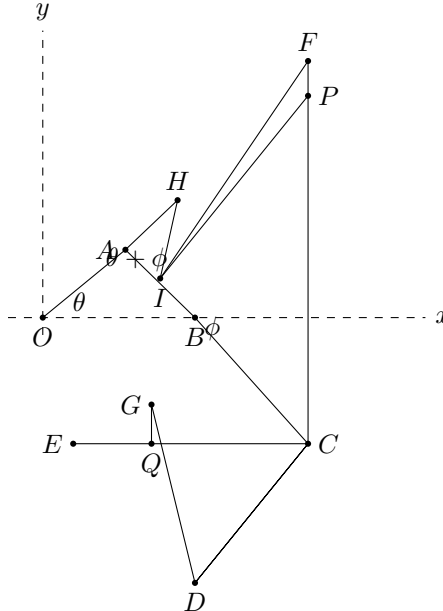
$$x = \sin \phi, \quad y = \frac{1 - \cos(\theta + \phi)}{\sin \phi} - \sin \theta.$$

I found, however, that this was not the cubic curve most easily constructed; and I ultimately devised a mechanical arrangement consisting of

1. Rod  $OH$ , and connected with it by a pin at  $H$ , rod  $HI$ .<sup>1</sup>

*p. 148 [source PDF p. 212]*

2. Square  $ACD$ , and connected with it by a pin at  $D$ , rod  $DG$ .
3. Square  $ECF$ ; the two squares being connected by a pin at  $C$ .
4. Rod  $IJ$ .



The rod  $OH$  rotates about a pin at  $O$ ; taking  $HA = HI$ , there is a pin at  $A$  connecting a fixed point of this rod with the extremity  $A$  of the square  $ACD$ : the fixed point  $B$  of this square moves along the line  $Ox$ . There is a pin at  $I$  connecting the extremities of the rods  $HI, IJ$ ; and this slides along the leg  $AC$  of the square  $ACD$ , the rod  $IJ$  being always at right angles thereto: finally the legs of the square  $ECF$  are always parallel to  $Ox, Oy$ , and the rod  $DG$  at right angles to  $EC$ . I have omitted from the description the parallel-motion rods or other arrangements necessary for giving

<sup>1</sup>There was a mechanical convenience in this, but observe that producing  $OH$  to meet  $IP$  in  $I'$ , the single straight rod  $OHI'$  might have been made use of.

these fixed directions to the rod  $IJ$ , the square  $ECF$ , and the rod  $DG$ . It will be seen that the angles  $AOB, ABO$  are variable angles connected by an equation of the form above referred to; and that the lines  $IJ, CF$  determine by their intersection the point  $P$ ; and the lines  $CE, DG$  determine by their intersection the point  $Q$ ; the curve about to be considered is that determined by the relative motion of  $P$  in regard to  $Q$ ; or say the curve the coordinates of a point of which are

$$x = QC, \quad y = CP.$$

I write

$$\begin{aligned} \angle AOB &= \theta, & \angle ABO &= \phi, \\ OA &= a, & AB &= b, & AC &= c, & CD &= d, & AH &= HI = \frac{1}{2}h. \end{aligned}$$

*p. 149 [source PDF p. 213]*

We then have  $a \sin \theta = b \sin \phi$ ; and moreover the length  $AI$  being  $= h \cos(\theta + \phi)$ , and therefore  $IC = c - h \cos(\theta + \phi)$ , we have

$$y = CP = \frac{c - h \cos(\theta + \phi)}{\sin \phi}, \quad x = QC = d \sin \phi;$$

whence also

$$xy = d\{c - h \cos(\theta + \phi)\};$$

or we have

$$xy = d \left\{ c - h \sqrt{1 - \frac{x^2}{d^2}} \sqrt{1 - \frac{b^2 x^2}{a^2 d^2}} + h \frac{b x^2}{a d^2} \right\},$$

that is

$$x \left( y - \frac{bh}{ad} x \right) - cd = -dh \sqrt{1 - \frac{x^2}{d^2}} \sqrt{1 - \frac{b^2 x^2}{a^2 d^2}};$$

or rationalising and reducing, this is

$$x^2 y^2 - \frac{2bh}{ad} x^3 y - 2cdxy \left\{ 2 \frac{b}{a} ch + h^2 \left( 1 + \frac{b^2}{a^2} \right) \right\} x^2 d^2 (c^2 - h^2) = 0,$$

a quartic curve with two dps.

In the particular case  $a = b$ , the relation between  $\theta, \phi$  is simply  $\theta = \phi$ ; the curve should become unicursal.

Writing in the equation  $\frac{b}{a} = 1$ , the equation takes the form

$$\left\{ x \left( y - \frac{2h}{d} x \right) - d(c - h) \right\} \{ xy - d(c + h) \} = 0;$$

the second factor is extraneous, and the curve is the hyperbola

$$x \left( y - \frac{2h}{d} x \right) - d(c - h) = 0,$$

as at once appears from the foregoing irrational form of the equation.

In the particular case  $h = c$ , the equation contains the factor  $x$ , and omitting this it becomes

$$x \left( y^2 - \frac{2bc}{ad} x^2 \right) - 2cdyc \left( 1 + \frac{b}{a} \right)^2 x = 0;$$

viz. we have here a cubic curve with three real asymptotes meeting in a point which is also the centre of the curve.

If simultaneously  $a = b$  and  $h = c$ , then the equation is

$$x \left( y - \frac{2c}{d}x \right) (xy - 2cd) = 0,$$

the actual locus being in this case the line  $y - \frac{2c}{d}x = 0$ .

*p. 150 [source PDF p. 214]*

Writing  $h = c$ , and for greater convenience  $h = c = d = 1$ ; also to fix the ideas supposing  $b < a$ , and writing  $\frac{b}{a} = k, = \sin \lambda$ , then we have

$$\sin \theta = k \sin \phi, \quad x = \sin \phi, \quad y = \frac{1 - \cos(\theta + \phi)}{\sin \phi};$$

that is

$$xy = 1 - \sqrt{1 - x^2} \sqrt{1 - k^2 x^2} + kx^2,$$

giving the rationalised equation

$$x(y^2 - 2kx^2) - 2y + 4x = 0;$$

the angle  $\phi$  may be anything whatever, but  $\theta$  varies between the limits  $\pm\lambda$ , the simultaneous values of these angles and of the coordinates being

|                    |                     |          |                           |
|--------------------|---------------------|----------|---------------------------|
| $\phi = 0$         | $\theta = 0$        | $x = 0$  | $y = 0$                   |
| $\phi = 90^\circ$  | $\theta = \lambda$  | $x = 1$  | $y = 1 + \sin \lambda$    |
| $\phi = 180^\circ$ | $\theta = 0$        | $x = 0$  | $y = \pm\infty$           |
| $\phi = 270^\circ$ | $\theta = -\lambda$ | $x = -1$ | $y = -(1 + \sin \lambda)$ |
| $\phi = 360^\circ$ | $\theta = 0$        | $x = 0$  | $y = 0$                   |

and it thus appears that the mechanism gives the continuous branch which belongs to the asymptote  $x = 0$  of the cubic curve; the other two branches belong to

$$x = \sin \phi, \quad y = \frac{1 + \cos(\theta + \phi)}{\sin \phi},$$

which would require a slight alteration in the arrangement of the mechanism.

I remark that if  $AH, HI$  had been unequal, then writing  $\angle HIA = \chi$ , this would be connected with  $\theta + \phi$  by an equation of the form

$$\sin(\theta + \phi) = m \sin \chi,$$

and the coordinates  $x, y$  would be rational functions of the sines and cosines of  $\theta, \phi, \chi$ ; the deficiency is in this case  $> 1$ .

*p. 151 [source PDF p. 215]*

507.

## ON THE MECHANICAL DESCRIPTION OF CERTAIN QUARTIC CURVES BY A MODIFIED OVAL CHUCK

*[From the Proceedings of the London Mathematical Society, vol. iv. (1871–1873), pp. 186–190. Read December 12, 1872.]*

The geometrical principle of the oval chuck is the well-known one that if a plane move in such manner that two lines  $Ox, Oy$ , fixed in the plane and moveable with it, pass through two fixed points  $A, B$ , respectively, then any fixed point  $P$  traces out on the plane an ellipse. The point  $A$

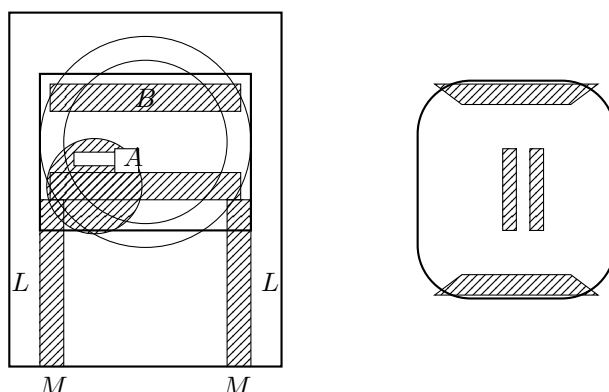
is on the (geometrical) axis of the mandril; there is connected with the head a guide-ring moving horizontally; the point  $B$  is the centre of the guide-ring, this being a ring connected with the head, moveable horizontally at right angles to the axis in such wise that the distance  $AB$  of the two centres is adjustable to any given value; the fixed point  $P$  is the tool, which practically is held on the level of the axis, that is, at a point in the line  $AB$ . The guide-ring remains fixed during the motion of the lathe.

It occurred to me that a chuck applicable to ornamental turning might be constructed by giving to the guide-ring a reciprocating motion synchronous with the rotation of the mandril; viz. for this purpose it is only necessary to affix to the axis of the mandril an eccentric, working in a frame attached to the guide-ring so as to move the centre  $B$  of the guide-ring backwards and forwards along the line  $AB$ : the curve is thus that described by the fixed point  $P$  upon a plane moving in such manner that the lines  $Ox, Oy$  pass always through the points  $A, B$  respectively; the former of these being a fixed point, the latter of them a point moving according to determinate law backwards and forwards along a fixed line through  $A$ .

The plan is carried out in a drawing apparatus which I have had constructed in wood, the axis being here vertical instead of horizontal, and the details of course different from what they would be for a lathe.

*p. 152 [source PDF p. 216]*

The apparatus consists of a piece of inch-board, about 10 inches long by 7 inches broad, pierced with a circular hole of 1 inch diameter for a vertical axis: the edges of the board serve as guides for the frame  $L$ , which carries the guide-ring, and resting on the board we have the frame  $M$ , itself guided by the frame  $L$ : the two frames move independently of each other, but they can be clamped together; the axis has



upon it a square nut, the sides of which work in the slot of an eccentric, the throw of this being adjustable by means of a screw passing through the nut and axis, and there is above the eccentric a square nut shown in the figure. This is capable of rotation round the axis, so that two of its sides may be placed either parallel with or inclined to the sides of the slot; but I fix it with two sides parallel to those of the slot by means of a screw run into the axis. The upper surfaces of the last-mentioned nut and of the guide-ring are flush with each other; and we then have a table or bed having, on its under-surface, guides which work on the outer edges of the guide-ring and on two edges of the nut. It will be observed that the bed may be placed in two different positions, viz. the guides may work on either pair of edges of the nut, those which are parallel to the sides of the slot, or those which are at right angles to it.

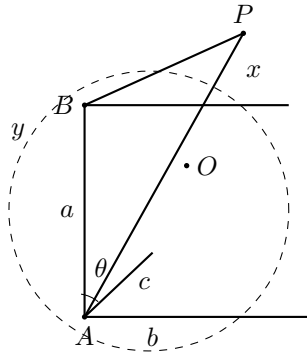
Supposing the bed placed as above upon the guide-ring and nut, then if the frames  $L$  and  $M$  are disconnected, and the former of them is fixed, the frame  $M$  will, on rotation of the axis, be carried backwards and forwards by the eccentric, but this will in no wise affect the motion of the bed; the arrangement is then equivalent to the oval chuck, and a pencil fixed above the bed in any given position will trace out upon it an ellipse. If, however, the frame  $L$ , instead of being fixed, is clamped

to the frame  $M$ , then the two frames, and therefore the guide-ring, are carried backwards and forwards by the eccentric, and the curve traced out by the pencil is no longer an ellipse; it is, as I proceed to show, a special form of trinodal quartic; viz. there is a tacnode (= two nodes) at infinity, and a third node, which may be a crunode, cusp, or acnode. In the last-mentioned case, the acnode or conjugate point is, as usual, not exhibited by the mechanical description, and the curve has no visible singularity.

*p. 153 [source PDF p. 217]*

Let the coordinates of the fixed point  $P$ , referred to axes through  $A$ , the first of them perpendicular to, and the second coincident with,  $AB$ , be  $b, c$ ; let the distance  $AB$  be  $= a$ ; and let  $\theta$  denote the angle  $BAO$ : then, if  $x, y$  are the coordinates of  $P$  referred to the origin  $O$  and axes  $Ox, Oy$ , we have

$$x + a \cos \theta = b \sin \theta + c \cos \theta, \quad y = -b \cos \theta + c \sin \theta,$$



which, if  $a$  be constant, gives a quadric equation, or the curve is an ellipse; and, in particular, if  $b = 0$ , that is if the point  $P$  is on the line  $AB$ , then we have

$$x = (c - a) \cos \theta, \quad y = c \sin \theta,$$

or the curve is

$$\frac{x^2}{(c - a)^2} + \frac{y^2}{c^2} = 1.$$

But if  $a$  is a given function of  $\theta$ , then the equation is still found by eliminating  $\theta$  between the two equations for  $x$  and  $y$ . In particular, if the distance  $AB$  is given as the perpendicular upon the tangent of a circle, as shown in the figure, then if  $k$  be the radius  $AC$  of this circle, and  $\lambda$  the inclination of  $AC$  to  $Ax$  ( $k$  and  $\lambda$  being taken to be constants), we have

$$a = k \cos(\theta + \lambda),$$

and the equations are

$$x = b \sin \theta + \{c - k \cos(\theta + \lambda)\} \cos \theta, \quad y = -b \cos \theta + c \sin \theta.$$

The elimination is nearly the same as if  $b$  were  $= 0$ ; viz. we may determine  $\gamma, \alpha$  in such wise that

$$\begin{aligned} b \sin \theta + c \cos \theta &= \gamma \cos(\theta + \alpha), = \gamma \cos \phi \text{ suppose,} \\ -b \cos \theta + c \sin \theta &= \gamma \sin(\theta + \alpha), = \gamma \sin \phi; \end{aligned}$$

*p. 154 [source PDF p. 218]*

and then

$$x = \gamma \cos \phi - k \cos(\phi + \lambda - \alpha) \cos(\phi - \alpha), \quad y = \gamma \sin \phi.$$

I will, for greater simplicity, at once write  $b = 0$ : the equations thus are

$$x = (c - k \cos \theta \cos \lambda + k \sin \theta \sin \lambda) \cos \theta, \quad y = c \sin \theta;$$

or say the first is

$$x = (c + k \sin \theta \sin \lambda) \cos \theta - k \cos^2 \theta \cos \lambda;$$

whence we have

$$x + \frac{k \cos \lambda}{c^2}(c^2 - y^2) = \left(c + \frac{ky}{c} \sin \lambda\right) \frac{1}{c} \sqrt{c^2 - y^2},$$

that is

$$c^2 x + k \cos \lambda (c^2 - y^2) = (c^2 + ky \sin \lambda) \sqrt{c^2 - y^2},$$

an equation which will assume a more simple form if either  $\lambda = 0$  or  $\lambda = 90^\circ$ ; that is, if in the apparatus the nut-sides which guide the bed are either at right angles, or parallel, to the sides of the slot.

Taking the general case, and writing for convenience  $\cos \theta = \xi$ ,  $\sin \theta = \eta$ , the curve is given by equations of the form

$$x = (\xi, \eta, 1)^2, \quad y = (\xi, \eta, 1)^2, \quad \xi^2 + \eta^2 = 1;$$

viz. the elimination of  $\xi, \eta$  from these equations leads to the equation of the curve. The points of the curve have thus a  $(1, 1)$  correspondence with those of the circle  $\xi^2 + \eta^2 = 1$ ; or, the circle being unicursal, the curve is also unicursal. Moreover, considering the intersections of the curve with an arbitrary line  $ax + by + c = 0$ , the points of intersection correspond to the points of intersection of the circle by the quadric

$$a(\xi, \eta, 1)^2 + b(\xi, \eta, 1)^2 + c = 0;$$

viz. there are four points of intersection, or the curve is a quartic, and hence it is a binodal quartic. But it is a binodal quartic of a special form: to show this more clearly, I introduce for homogeneity the coordinates  $z, \zeta$ , so that the foregoing equations become

$$x : y : z = (\xi, \eta, \zeta)^2 : (\xi, \eta, \zeta)^2 : \zeta^2, \quad \text{where } \xi^2 + \eta^2 - \zeta^2 = 0;$$

the curve corresponding to these equations is, as just seen, a binodal quartic. But in the case in hand the form is the more special one,

$$x : y : z = (\xi, \eta, \zeta)^2 : \xi \zeta : \zeta^2, \quad \text{where } \xi^2 + \eta^2 - \zeta^2 = 0.$$

The intersections by the arbitrary line  $ax + by + cz = 0$  are the points corresponding to the intersections of the circle  $\xi^2 + \eta^2 - \zeta^2 = 0$  by the quadric

$$a(\xi, \eta, \zeta)^2 + b\xi\zeta + c\zeta^2 = 0,$$

*p. 155 [source PDF p. 219]*

giving four intersections. But the intersections by the line  $by + cz = 0$  (that is, by any line through the point  $y = 0, z = 0$ ) are obtained from the equation  $b\xi\zeta + c\zeta^2 = 0$ ; viz. this breaks up into  $b\xi + c\zeta = 0$ ,  $\zeta = 0$ , and the last factor combined with the equation of the circle gives  $\zeta = 0$ ,  $\xi^2 + \eta^2 = 0$ , the two circular points at infinity, corresponding each to the point  $y = 0, z = 0$ : the other factor gives points corresponding to two variable points on the curve; that is, a line through the point  $y = 0, z = 0$  meets the curve in this point twice and in two other points. Again, making  $b = 0$ , or taking the line to be the line at infinity  $z = 0$ , the equations then are  $\zeta = 0$ ,  $\xi^2 + \eta^2 = 0$ ; viz. we then have the circular points at infinity each twice, corresponding to the point  $y = 0, z = 0$  four times, and no other point; that is, the line  $z = 0$  meets the curve in the point  $y = 0, z = 0$  four times. We thus see that the curve has at  $y = 0, z = 0$ , that is at infinity on the line  $y = 0$ , a tacnode (counting as two nodes), the tangent at this point being the line at infinity  $z = 0$ . The curve being trinodal has of course one other node.

*p. 156 [source PDF p. 220]*



## ON GEODESIC LINES, IN PARTICULAR THOSE OF A QUADRIC SURFACE

[From the Proceedings of the London Mathematical Society, vol. iv. (1871–1873), pp. 191–211. Read December 12, 1872.]

The present Memoir contains an investigation of the differential equation (of the second order) of the geodesic lines on a surface, the coordinates of a point on the surface being regarded as given functions of two parameters  $p, q$ , and researches in connection therewith; a deduction of Jacobi's differential equation of the first order in the case of a quadric surface, the parameters  $p, q$  being those which determine the two sets of curves of curvature; formulae where the parameters are those which determine the two right lines through the surface; and a discussion of the forms of the geodesic lines in the two cases of an ellipsoid and a skew hyperboloid respectively.

## Preliminary Formulae.

1. I call to mind the fundamental formulae in the Memoir by Gauss, “Disquisitiones generales circa superficies curvas,” *Comm. Gott. recent.*, t. vi., 1827, (reprinted as an Appendix in Liouville's edition of Monge,) together with some that I have added to them. The coordinates  $x, y, z$  of a point on a surface are regarded as given functions of two parameters  $p, q$ , these expressions of  $x, y, z$  in effect determining the equation of the surface, and we have

$$\begin{aligned} dx + \frac{1}{2}d^2x &= a dp + a' dq + \frac{1}{2}(\alpha dp^2 + 2\alpha' dp dq + \alpha'' dq^2), \\ dy + \frac{1}{2}d^2y &= b dp + b' dq + \frac{1}{2}(\beta dp^2 + 2\beta' dp dq + \beta'' dq^2), \\ dz + \frac{1}{2}d^2z &= c dp + c' dq + \frac{1}{2}(\gamma dp^2 + 2\gamma' dp dq + \gamma'' dq^2), \\ A, B, C &= bc' - b'c, \quad ca' - c'a, \quad ab' - a'b; \end{aligned}$$

*p. 157 [source PDF p. 221]*

whence differential equation of surface is

$$Adx + Bdy + Cdz = 0.$$

Also

$$E, F, G = a^2 + b^2 + c^2, \quad aa' + bb' + cc', \quad a'^2 + b'^2 + c'^2;$$

so that element of length on the surface is given by

$$dx^2 + dy^2 + dz^2 = E dp^2 + 2F dp dq + G dq^2;$$

or, as I write it,

$$= (E, F, G)(dp, dq)^2;$$

and moreover

$$V^2 = A^2 + B^2 + C^2 = EG - F^2.$$

The equation  $(E, F, G)(dp, dq)^2 = 0$  determines at each point on the surface two directions (necessarily imaginary) which are called the “circular” directions. Passing on the surface from point to point along the circular directions, we obtain two series of curves (always imaginary) which are the “circular” curves; the equation  $(E, F, G)(dp, dq)^2 = 0$  is the differential equation of these curves; and if we have  $E = 0, G = 0$ , then this becomes  $dp dq = 0$ ; viz. we have in this case  $p = \text{const.}$  and  $q = \text{const.}$  as the equations of the two sets of circular curves respectively. It is clear *a priori*, and will be shown analytically in the sequel, that the circular curves are geodesic lines.

I write also

$$E', F', G' = A\alpha + B\beta + C\gamma, \quad Aa' + B\beta' + C\gamma', \quad Aa'' + B\beta'' + C\gamma'',$$

or, what is the same thing,  $E', F', G'$  represent the determinants

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ \alpha & \beta & \gamma \end{vmatrix}, \quad \begin{vmatrix} a & b & c \\ a' & b' & c' \\ \alpha' & \beta' & \gamma' \end{vmatrix}, \quad \begin{vmatrix} a & b & c \\ a' & b' & c' \\ \alpha'' & \beta'' & \gamma'' \end{vmatrix},$$

respectively, (these last symbols do not occur in Gauss). [They are the  $D, D', D''$  of Gauss.]

2. The radius of curvature of normal section corresponding to direction  $dp : dq$  is given by

$$\frac{\rho}{V} = \frac{(E, F, G)(dp, dq)^2}{(E', F', G')(dp, dq)^2};$$

whence it appears that the directions of the inflexional or chief tangents (the *Haupttangenten*) are determined by the equation

$$(E', F', G')(dp, dq)^2 = 0.$$

The directions in question are imaginary on a surface such as the ellipsoid where the curvatures are in the same direction, but on a concavo-convex surface they are real; and in particular on the hyperboloid they coincide with the directions of the generating lines. We may on any surface pass from point to point along the chief directions; we have thus on the surface two sets of curves which are the chief curves;

*p.* 158 [source PDF *p.* 222]

the differential equation of these is  $(E', F', G')(dp, dq)^2 = 0$ ; and in particular if  $E' = 0$ ,  $G' = 0$ , then this becomes  $dp dq = 0$ , or we have  $p = \text{const.}$ ,  $q = \text{const.}$  for the equations of the two sets of chief curves respectively. On the hyperboloid the chief curves are the two sets of generating lines. The chief curves are not in general geodesic lines, but on the hyperboloid, quà straight lines, they are, it is clear, geodesic lines.

3. The directions of the curves of curvature, or the principal tangents, and the corresponding values of the radius of curvature are determined by

$$\frac{\rho}{V} = \frac{Edp + Fdq}{E'dp + F'dq} = \frac{Fdp + Gdq}{F'dp + G'dq};$$

or, what is the same thing, these directions are determined by the equation

$$\begin{vmatrix} dq^2 & -dq dp & dp^2 \\ E & F & G \\ E' & F' & G' \end{vmatrix} = 0.$$

The same equations may be written

$$-dq : dp = \frac{\rho E' - VE}{\rho F' - VF} = \frac{\rho F' - VF}{\rho G' - VG};$$

that is the principal radii of curvature are determined by the equation

$$\rho^2(E'G' - F'^2) - \rho V(EG' + E'G - 2FF') + V^2(EG - F^2) = 0,$$

(last term is  $= V^4$ , but it is better to retain the original form): and then,  $\rho$  being either root, the last preceding equations give the direction of the curve of curvature corresponding to the given value of the radius of curvature.

If  $p = \text{const.}$ ,  $q = \text{const.}$  are the equations of the two systems of curves of curvature respectively, then the quadric equation in  $(dp, dq)$  must become  $dp dq = 0$ ; this will be so if  $F = 0$ ,  $F' = 0$ ; and we thus have these equations, viz. written at full length they are

$$d_p x d_q x + d_p y d_q y + d_p z d_q z = 0,$$

$$\begin{vmatrix} d_p x & d_p y & d_p z \\ d_q x & d_q y & d_q z \\ d_p d_q x & d_p d_q y & d_p d_q z \end{vmatrix} = 0,$$

as the conditions in order that  $p = \text{const.}$ ,  $q = \text{const.}$  may be the two systems of curves of curvature. The former of these equations merely expresses that the two sets of curves always intersect at right angles.

*p. 159 [source PDF p. 223]*

### *General Theory of the Geodesic Lines on a Surface.*

4. I now proceed to investigate the theory of geodesic lines on a surface, the surface being determined as above by means of given expressions of the coordinates  $x, y, z$  in terms of the parameters  $p, q$ .

The differential equation obtained by Gauss for the geodesic lines is in a form not symmetrical in regard to the two variables; viz. his equation is

$$\frac{dE}{dq} dp^2 + 2 \frac{dF}{dp} dp dq + \frac{dG}{dp} dq^2 = 2 ds d \frac{E dp + F dq}{ds},$$

where, as above,

$$ds^2 = (E, F, G)(dp, dq)^2.$$

If we herein consider  $p, q$  as functions of a parameter  $\theta$ , and write for shortness

$$d_\theta p, d_\theta q, d_\theta^2 p, \&c. = p', q', p'', \&c.,$$

also

$$\Omega = (E, F, G)(p', q')^2,$$

and

$$d_p E = E_1, \quad d_q E = E_2, \quad \&c.,$$

then the equation is

$$(E_1, F_1, G_1)(p', q')^2 - 2\sqrt{\Omega} \left( \frac{Ep' + Fq'}{\sqrt{\Omega}} \right)' = 0.$$

We have

$$\left( \frac{Ep' + Fq'}{\sqrt{\Omega}} \right)' = \frac{1}{\Omega\sqrt{\Omega}}(M + N),$$

where  $N$  is the part containing  $p'', q''$ , which I will first calculate; viz. we have

$$\begin{aligned} N &= \Omega(Ep'' + Fq'') - (Ep' + Fq')\frac{1}{2}\Omega', \\ &= \Omega(Ep'' + Fq'') - (Ep' + Fq')\{(Ep' + Fq')p'' + (Fp' + Gq')q''\}, \\ &= p''\{E\Omega - (Ep' + Fq')^2\} + q''\{F\Omega - (Ep' + Fq')(Fp' + Gq')\}; \end{aligned}$$

or substituting for  $\Omega$  its value, this is

$$= p''(EG - F^2)q'^2 - q''(EG - F^2)p'q', = -q'(EG - F^2)(p'q'' - p''q');$$

wherefore

$$\left( \frac{Ep' + Fq'}{\sqrt{\Omega}} \right)' = \frac{1}{\Omega\sqrt{\Omega}}\{M - q'(EG - F^2)(p'q'' - p''q')\},$$

and the equation becomes

$$\Omega(E_1 p'^2 + 2F_1 p' q' + G_1 q'^2) - 2M + 2q'(EG - F^2)(p' q'' - p'' q') = 0;$$

whence we foresee that the whole equation must divide by  $q'$ .

*p. 160 [source PDF p. 224]*

5. We have

$$\begin{aligned} M &= (p' dE + q' dF) \Omega - (Ep' + Fq')(\tfrac{1}{2} p'^2 dE + p' q' dF + \tfrac{1}{2} q'^2 dG), \\ &= dE \{ p' \Omega - \tfrac{1}{2} p'^2 (Ep' + Fq') \} \\ &\quad + dF \{ q' \Omega - p' q' (Ep' + Fq') \} \\ &\quad + dG \{ -\tfrac{1}{2} q'^2 (Ep' + Fq') \}, \end{aligned}$$

or say

$$\begin{aligned} 2M &= dE[p'^2(p'E + q'F) + 2p'q'(p'F + q'G)] \\ &\quad + dF[2q'^2(p'F + q'G)] \\ &\quad + dG[-q'^2(p'E + q'F)], \\ &= (E_1 p' + E_2 q')[p'^2(Ep' + Fq') + 2p'q'(Fp' + Gq')] \\ &\quad + (F_1 p' + F_2 q')[2q'^2(Fp' + Gq')] \\ &\quad + (G_1 p' + G_2 q')[-q'^2(Ep' + Fq')]. \end{aligned}$$

The term in  $p'^4$  is  $EE_1 p'^4$ , which is also the term in  $p'^4$  of  $\Omega(E_1 p'^2 + 2F_1 p' q' + G_1 q'^2)$ ; whence  $\Omega(E_1 p'^2 + 2F_1 p' q' + G_1 q'^2) - 2M$  divides by  $q'$ .

Proceeding to the reduction:

Term in  $E_1$  is

$$E_1 \cdot p'^2 \Omega - p'^3 (Ep' + Fq') - 2p'^2 q' (Fp' + Gq') = E_1 \cdot (-p'^2 q' (Fp' + Gq'));$$

term in  $F_1$  is

$$F_1 \cdot 2p' q' \Omega - 2p' q'^2 (Fp' + Gq') = F_1 \cdot 2p'^2 q' (Ep' + Fq');$$

term in  $G_1$  is

$$G_1 \cdot q'^2 \Omega + p' q'^2 (Ep' + Fq') = G_1 \{ 2p' q'^2 (Ep' + Fq') + q'^3 (Fp' + Gq') \}.$$

6. The remaining terms in  $E_2, F_2, G_2$  require no reduction, and the result is

$$\begin{aligned} &E_1 \{ -p'^2 (Fp' + Gq') \} - E_2 \{ p'^2 (Ep' + Fq') + 2p' q' (Fp' + Gq') \} \\ &+ F_1 \{ 2p'^2 (Ep' + Fq') \} - F_2 \{ 2q'^2 (Fp' + Gq') \} \\ &+ G_1 \{ 2p' q' (Ep' + Fq') + q'^2 (Fp' + Gq') \} - G_2 \{ -q'^2 (Ep' + Fq') \} \\ &+ 2(EG - F^2)(p' q'' - p'' q') = 0, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} &(Ep' + Fq') \{ (2F_1 - E_2) p'^2 + 2G_1 p' q' + G_2 q'^2 \} \\ &- (Fp' + Gq') \{ E_1 p'^2 + 2E_2 p' q' + (2F_2 - G_1) q'^2 \} \\ &+ 2(EG - F^2)(p' q'' - p'' q') = 0, \end{aligned}$$

which is the required differential equation of the second order: the independent variable has been taken to be the arbitrary quantity  $\theta$ ; but taking  $\theta = p$ , or  $q$ , say  $\theta = p$ , we have  $p' = 1$ ,  $p'' = 0$ , and the equation then contains (besides  $p$  and  $q$ ) only  $q''$  and  $q'$ , that is,  $d_p^2 q$  and  $d_p q$ , and is therefore a differential relation between  $p$  and  $q$ .

*p. 161 [source PDF p. 225]*

7. Instead of starting, as above, from the equation given by Gauss, we may use the geometrical property that at each point of a geodesic line the osculating plane passes through the normal of the surface.

Considering, as above,  $p, q$  as functions of a variable  $\theta$ , then,  $\theta$  becoming  $\theta + d\theta$ , the new values of  $p, q$  are

$$p + p'd\theta + \frac{1}{2}p''d\theta^2, \quad q + q'd\theta + \frac{1}{2}q''d\theta^2;$$

and that of  $x$  is

$$x + a(p'd\theta + \frac{1}{2}p''d\theta^2) + a'(q'd\theta + \frac{1}{2}q''d\theta^2) + \frac{1}{2}(\alpha p'^2 + 2\alpha'p'q' + \alpha''q'^2)d\theta^2;$$

or calling this  $x + x'd\theta + \frac{1}{2}x''d\theta^2$ , we have

$$x' = ap' + a'q', \quad x'' = ap'' + a'q'' + \alpha p'^2 + 2\alpha'p'q' + \alpha''q'^2,$$

and so

$$\begin{aligned} y' &= bp' + b'q', & y'' &= bp'' + b'q'' + \beta p'^2 + 2\beta'p'q' + \beta''q'^2, \\ z' &= cp' + c'q', & z'' &= cp'' + c'q'' + \gamma p'^2 + 2\gamma'p'q' + \gamma''q'^2. \end{aligned}$$

The condition in question is expressed by the equation

$$\begin{vmatrix} A & B & C \\ x' & y' & z' \\ x'' & y'' & z'' \end{vmatrix} = 0,$$

that is

$$\begin{aligned} & \begin{vmatrix} A & B & C \\ ap' + a'q' & bp' + b'q' & cp' + c'q' \\ ap'' + a'q'' & bp'' + b'q'' & cp'' + c'q'' \end{vmatrix} \\ & + \begin{vmatrix} A & B & C \\ ap' + a'q' & bp' + b'q' & cp' + c'q' \\ \alpha p'^2 + 2\alpha'p'q' + \alpha''q'^2 & \beta p'^2 + 2\beta'p'q' + \beta''q'^2 & \gamma p'^2 + 2\gamma'p'q' + \gamma''q'^2 \end{vmatrix} = 0. \end{aligned}$$

8. The first determinant is the sum of three terms such as  $A(bc' - b'c)(p'q'' - p''q')$ ; viz. this is  $A^2(p'q'' - p''q')$ , or the determinant is

$$(A^2 + B^2 + C^2)(p'q'' - p''q') = (EG - F^2)(p'q'' - p''q').$$

The second determinant is the sum of three terms such as

$$(\alpha p'^2 + 2\alpha'p'q' + \alpha''q'^2)[B(cp' + c'q') - C(bp' + b'q')],$$

where the factor in [ ] is

$$p'[c(ca' - c'a) - b(ab' - a'b)] + q'[c'(ca' - c'a) - b'(ab' - a'b)],$$

*p. 162 [source PDF p. 226]*

which is  $= p'(a'E - aF) + q'(a'F - aG)$ . We have thus the second determinant, and the equation becomes

$$\begin{aligned} & (EG - F^2)(p'q'' - p''q') \\ & + (\alpha p'^2 + 2\alpha'p'q' + \alpha''q'^2)\{p'(a'E - aF) + q'(a'F - aG)\} \\ & + (\beta p'^2 + 2\beta'p'q' + \beta''q'^2)\{p'(b'E - bF) + q'(b'F - bG)\} \\ & + (\gamma p'^2 + 2\gamma'p'q' + \gamma''q'^2)\{p'(c'E - cF) + q'(c'F - cG)\} = 0, \end{aligned}$$

an equation of the same form as that previously obtained, and which can be without difficulty identified therewith.

*The Circular Curves are Geodesics.*

9. I proceed to show that the circular curves are geodesics; viz. that an integral of the geodesic equation is

$$(E, F, G)(p', q')^2 = 0.$$

Starting from this equation, we have

$$\begin{aligned} & 2\{(Ep' + Fq')p'' + (Fp' + Gq')q''\} \\ & + (E_1p' + E_2q')p'^2 + 2(F_1p' + F_2q')p'q' + (G_1p' + G_2q')q'^2 = 0. \end{aligned}$$

Now the equation, writing therein  $Ep' + Fq' = \lambda q'$ , gives  $Fp' + Gq' = -\lambda p'$ : these equations may be written

$$Ep' + (F - \lambda)q' = 0, \quad (F + \lambda)p' + Gq' = 0,$$

the value of  $\lambda$  being therefore  $-\lambda^2 = EG - F^2$ . The result just obtained thus becomes

$$\begin{aligned} & 2\lambda(p''q' - p'q'') \\ & + [(E_1p' + E_2q')p' + (F_1p' + F_2q')q'] \cdot \left(-\frac{1}{\lambda}\right) (Fp' + Gq') \\ & + [(F_1p' + F_2q')p' + (G_1p' + G_2q')q'] \cdot \frac{1}{\lambda} (Ep' + Fq'), \end{aligned}$$

that is

$$\begin{aligned} & 2(EG - F^2)(p'q'' - p''q') \\ & - (Fp' + Gq')[E_1p'^2 + (E_2 + F_1)p'q' + F_2q'^2] \\ & + (Ep' + Fq')[F_1p'^2 + (F_2 + G_1)p'q' + G_2q'^2] = 0; \end{aligned}$$

or what is the same thing, adding hereto the zero value

$$\Delta(E, F, G)(p', q')^2 = (Fp' + Gq')\Delta q' + (Ep' + Fq')\Delta p',$$

where  $\Delta$  is arbitrary, the equation is

$$\begin{aligned} & 2(EG - F^2)(p'q'' - p''q') \\ & + (Ep' + Fq')[F_1p'^2 + (F_2 + G_1)p'q' + G_2q'^2 + \Delta p'] \\ & - (Fp' + Gq')[E_1p'^2 + (E_2 + F_1)p'q' + F_2q'^2 - \Delta q'] = 0; \end{aligned}$$

viz. taking  $\Delta = (F_1 - E_2)p' + (G_1 - F_2)q'$ , this agrees with the geodesic equation.

The foregoing integral,  $(E, F, G)(p', q')^2 = 0$ , is, I believe, a particular, not a singular, solution of the geodesic equation.

*p. 163 [source PDF p. 227]*

*The Chief Lines are not in general Geodesics.*

10. That the chief lines are not in general geodesics appears most readily as follows:

To find the condition in order that  $p = \text{const.}$  may be a geodesic, we write in the geodesic equation  $p' = 0$ ,  $p'' = 0$ : the equation thus becomes

$$Fq' \cdot G_2q'^2 - Gq'(2F_2 - G_1)q'^2 = 0;$$

that is we have

$$FG_2 - 2F_2G + GG_1 = 0$$

as the condition in order that  $p = \text{const.}$  may be a geodesic: the condition that it may be a chief curve is  $G' = 0$ , which is a different condition.

We have of course, in like manner,

$$2EF_1 - E_1F - EE_2 = 0$$

as the condition in order that  $q = \text{const.}$  may be a geodesic; and  $E' = 0$  as the condition that this may be a chief curve. If  $p = \text{const.}, q = \text{const.}$  are each of them at once a geodesic and a chief curve, then the four equations must all be satisfied, viz. we must have

$$FG_2 - 2F_2G + GG_1 = 0, \quad 2EF_1 - E_1F - EE_2 = 0, \quad G' = 0, \quad E' = 0.$$

*Special Form of the Geodesic Equation.*

11. In the case where the curves  $p = \text{const.}, q = \text{const.}$  intersect at right angles (and in particular when these are the curves of curvature), we have  $F = 0$ ; whence also  $F_1 = 0, F_2 = 0$ ; and the geodesic equation assumes the more simple form,

$$\begin{aligned} & Ep'(-E_2p'^2 + 2G_1p'q' + G_2q'^2) \\ & - Gq'(E_1p'^2 + 2E_2p'q' - G_1q'^2) \\ & + 2EG(p'q'' - p''q') = 0. \end{aligned}$$

[11a. In the case of a surface of revolution we have

$$x = p \cos q, \quad y = p \sin q, \quad z = P;$$

$E$  is of the form  $1 + P'^2$ ,  $P' = d_p P$ , where  $P$  is a function of  $p$  only, and we have

$$ds^2 = (1 + P'^2)dp^2 + p^2dq^2,$$

that is

$$E = 1 + P'^2, \quad E_1 = 2P'P'', \quad E_2 = 0, \quad F = 0, \quad G = p^2, \quad G_1 = 2p, \quad G_2 = 0;$$

*p. 164 [source PDF p. 228]*

hence the differential equation is

$$(1 + P'^2)\{p^2(p'q'' - p''q') + 2pp'^2q'\} - p^2p'^2q'P'P'' + p^3q'^3 = 0;$$

this has an integral

$$\frac{(1 + P'^2)p'^2}{p^4q'^2} + \frac{1}{p^2} = \frac{1}{C^2},$$

or say

$$C^2s'^2 = p^4q'^2$$

where

$$s'^2 = (1 + P'^2)p'^2 + p^2q'^2.$$

Writing here  $\rho, \psi$  for  $p, q$ , where  $\rho$  is the distance of the point from the axis, and  $\psi$  is the longitude reckoned from an arbitrary meridian, then the equation is

$$Cds = \rho^2 d\psi,$$

which is the equation given by Legendre, *Théorie des fonctions elliptiques*, t. I. p. 361. This may also be written

$$\frac{C}{\rho} = \cos \gamma$$

if  $\gamma$  be the inclination of the geodesic line to the parallel of latitude.]

*Geodesics on a Quadric Surface.*

12. In the case of a quadric surface

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1,$$

writing for shortness  $\alpha, \beta, \gamma = b - c, c - a, a - b$ , we may express the coordinates  $x, y, z$  in terms of two parameters  $p, q$  as follows:

$$\begin{aligned} -\beta\gamma x^2 &= a(a+p)(a+q), \\ -\gamma\alpha y^2 &= b(b+p)(b+q), \\ -\alpha\beta z^2 &= c(c+p)(c+q), \end{aligned}$$

where, in fact,  $p = \text{const.}, q = \text{const.}$  are the equations of the two sets of curves of curvature respectively. Writing moreover

$$P = \frac{p}{(a+p)(b+p)(c+p)}, \quad Q = \frac{q}{(a+q)(b+q)(c+q)},$$

we have

$$ds^2 = \frac{1}{4}(p-q)(Pdp^2 - Qdq^2),$$

that is

$$E, F, G = \frac{1}{4}(p-q)P, \quad 0, \quad \frac{1}{4}(q-p)Q;$$

and the geodesic equation becomes

$$\begin{aligned} &Pp'\{Pp'^2 - 2Qp'q' + (Q + (q-p)Q')q'^2\} \\ &+ Qq'\{P + (p-q)P'\}p'^2 - 2Pp'q' + Qq'^2\} \\ &- 2(p-q)PQ(p'q'' - p''q') = 0, \end{aligned}$$

*p. 165 [source PDF p. 229]*

where  $P', Q'$  stand for  $d_p P$  and  $d_q Q$  respectively; viz. this is

$$\begin{aligned} &p'^3 \cdot P^2 \\ &+ p'^2 q' \cdot \{-PQ + (p-q)P'Q\} \\ &+ p'q'^2 \cdot \{-PQ + (q-p)PQ'\} \\ &+ q'^3 \cdot Q^2 \\ &- 2(p-q)PQ(p'q'' - p''q') = 0. \end{aligned}$$

13. This has a first integral,

$$p' \sqrt{\frac{P}{\theta + p}} + q' \sqrt{\frac{Q}{\theta + q}} = 0,$$

where  $\theta$  is the constant of integration; or say this is

$$\theta(p'^2 P - q'^2 Q) + p'^2 q P - q'^2 p Q = 0;$$

viz. differentiating logarithmically, this gives

$$\frac{2p'p''P - 2q'q''Q + p'^3 P' - q'^3 Q'}{p'^2 P - q'^2 Q} = \frac{2p'p''qP - 2q'q''pQ + p'^2(qp'P' + q'P) - q'^2(pq'Q' + p'Q)}{p'^2 q P - q'^2 p Q},$$

which, multiplying out the denominators, is in fact the foregoing geodesic equation. To verify, consider first the part involving  $p'', q''$ : this is

$$(2p'p''P - 2q'q''Q)(p'^2 q P - q'^2 p Q) - (2p'p''qP - 2q'q''pQ)(p'^2 P - q'^2 Q),$$



which is

$$= 2p'p''P \cdot q'^2Q(q-p) - 2q'q''Q \cdot p'^2P(q-p),$$

that is

$$= 2(q-p)PQp'q'(p''q' - p'q''),$$

or say

$$= 2(p-q)PQp'q'(p'q'' - p''q').$$

We have next the part

$$(p'^3P' - q'^3Q')(p'^2qP - q'^2pQ) - \{p'^2(qp'P' + q'P) + q'^2(pq'Q' + p'Q)\}(p'^2P - q'^2Q),$$

which is readily found to be

$$= -p'q'\{p'^3P^2 + p'^2q'(-PQ + (p-q)P'Q)p'q'^2(-PQ + (q-p)PQ') + q'^3Q^2\},$$

and the equation is thus verified.

14. We have consequently

$$dp\sqrt{\frac{p}{(a+p)(b+p)(c+p)(\theta+p)}} + dq\sqrt{\frac{q}{(a+q)(b+q)(c+q)(\theta+q)}} = 0,$$

involving the arbitrary constant  $\theta$  as the differential equation of the first order of the geodesic lines on the quadric surface

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1;$$

the geodesic lines in question

*p. 166 [source PDF p. 230]*

all touch the curve of curvature determined by the parameter  $\theta$ , that is the curve which is the intersection of the surface by the confocal surface

$$\frac{x^2}{a+\theta} + \frac{y^2}{b+\theta} + \frac{z^2}{c+\theta} = 1.$$

15. In the particular case  $\theta = \infty$ , the equation becomes

$$Pdp^2 - Qdq^2 = 0,$$

that is

$$\frac{p dp^2}{(a+p)(b+p)(c+p)} - \frac{q dq^2}{(a+q)(b+q)(c+q)} = 0,$$

which is the differential equation of the circular curves on the surface.

16. The signification of the case  $\theta = 0$  is not at first sight so obvious. Supposing that  $\theta$  is first indefinitely small, and writing the equation in the form

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} - 1 - \theta \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) + \&c. = 0,$$

we have the series of geodesics touching the (imaginary) curve of curvature, the intersection of the surface by the imaginary cone

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0.$$

These are, in fact, the right lines on the surface: I apprehend that the intersection in question is not a proper envelope, but is the locus of nodes of the geodesics, viz. each geodesic is to be considered as a pair of lines belonging to the two sets: I do not, however, quite understand this.

17. I say that the geodesics in question are the right lines on the surface; viz. writing in the differential equation  $\theta = 0$ , it is to be shown that the differential equation of the right lines is

$$\frac{dp}{\sqrt{(a+p)(b+p)(c+p)}} + \frac{dq}{\sqrt{(a+q)(b+q)(c+q)}} = 0,$$

or what is the same thing, that the integral of this equation represents the right lines on the surface.

Writing the equation of the surface in the form

$$\frac{x^2}{a} + \frac{y^2}{b} = 1 - \frac{z^2}{c},$$

we have at once

$$\begin{aligned}\frac{x}{\sqrt{a}} + \frac{iy}{\sqrt{b}} &= \sigma \left( 1 + \frac{z}{\sqrt{c}} \right), \\ \frac{x}{\sqrt{a}} - \frac{iy}{\sqrt{b}} &= \frac{1}{\sigma} \left( 1 - \frac{z}{\sqrt{c}} \right).\end{aligned}$$